

GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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ABSTRACT. A prepared causal experiment generates a predictive quotient and an actual probability law on it. We connect the geometry of this law to implementable finite-memory prediction and to resource-bounded comparison of experiments. The central theorem combines anisotropic acquired-mass estimates with an innovation-localized error process. A positive-kernel resolvent derives its second-moment bound from a one-step Lyapunov inequality, allowing recurrent expanding updates and arbitrarily long exact waits. The resulting checkpoint and online bounds match on the certified class; finite intersecting strata and disappearing widths are included. Causal morphisms transfer a common bounded loss while retaining acquisition, persistent state, workspace, program, precision, and physical-time accounts. A continuation-separation theorem proves a matching total-state lower bound with growing mandatory interface memory. A single explicitly specified robust experiment simultaneously realizes acquisition, geometric resolution, calibration, numerical-information, and optimal-deficiency terms in one minimax risk. Three raw-kernel verifications distinguish Bayesian filtering, singular absorbing acquisition, and recurrent fold-and-refresh dynamics. The geometric and moment arguments precede and do not depend on these realizations.

1. THE EXPERIMENTAL PROBLEM

The mother problem is to pass from a physically executable experiment to a quantitative law for the predictions that can actually be retained. There are two distinct obstructions. A set of formally reachable predictions need not receive appreciable mass under the preparation and exploration being used. Moreover, a good codebook at a checkpoint need not admit a good recursive implementation. The first obstruction is probabilistic; the second is causal. Treating either as a dimension count loses the mechanism of the problem.

Our main theorem connects these obstructions by weighting the propagation of quantization errors with the same acquisition probabilities that enter the lower bound. Inactive steps are especially important. When an update preserves a retained representative exactly, rounding need not be repeated. Charging an error at every physical tick would produce a spurious inverse acquisition probability in a system that simply waits. The theorem retains the actual clock while localizing error injection to the updates that require it.

The principal mechanism has two parts. A positive-kernel resolvent computes the energy of past rounding innovations from one-step experimental data. Its Lyapunov hypothesis permits expansion, rather than assuming that every informative update contracts. A separate continuation-separation argument decides when additional interface states are genuinely mandatory. This prevents a Cartesian-product implementation bound from masquerading as a memory lower bound. These mechanisms yield, respectively, a general causal transfer theorem and a common-task robust minimax theorem with growing total-state overhead.

Predictive states and their minimality have classical precedents in causal-state theory and controlled predictive representations [9, 5]. Conditional projection and probability quantization are also classical [4], as are finite filter approximation [6, 7, 3] and comparison of experiments [1, 13, 14].

Date: October 7, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 60G35; Secondary 94A29, 28A80, 93E11.

Key words and phrases. causal experiment, predictive quotient, acquired law, finite memory, singular quantization, experiment comparison.

Positive-operator moment methods are related to mean-square stability of Markov jump systems [12]. The contribution asserted here is the connection of these mechanisms to actual acquisition mass, forced causal approximation, and a precisely specified resource cut. The theorem-level distinctions are given in Section 8; no priority is claimed for the component theories.

1.1. Raw data, strategies, and preparation. A discrete-time experiment consists of standard Borel spaces $X_t, \mathcal{A}_t, Y_{t+1}$, a probability preparation μ on X_0 , and nonnegative normalized kernels

$$(1) \quad K_t(dx', dy, dc \mid x, a).$$

The visible history h_t records actions, reports, and the published nonnegative resource increments c . These increments include failed preparations and physical elapsed time. Positivity means nonnegativity and normalization, not a positive lower bound for every report. Deterministic transitions are permitted. An unknown physical parameter, when present, is included in the hidden state with a specified prior for the Bayesian assertions below. Parameter-uniform experiment comparison will be stated separately.

A legal strategy is a stochastic kernel $\alpha_t(da \mid h_t)$ supported on the declared available actions. All randomness used by the apparatus or algorithm is independent of the hidden preparation unless generated by (1). A stopping rule is a stopping time for the visible filtration. The main risk theorem fixes a deterministic raw-acquisition horizon N and an exploration strategy β independently of the encoder. It does not optimize β . At most N actual calls, including unsuccessful calls, are made. A stopped comparison is always explicitly bounded by N .

Successive kernel integration gives the path law P^α . The complete conditional belief $\pi_t(h_t)$ on the hidden state has a measurable version. We specify versions on a declared admissible history domain; identities on impossible reports are not inferred by division by zero.

1.2. Executable future tests and the quotient. A future test is a legal finite continuation followed by a bounded measurable function of its actual reports. Hidden-state functions qualify only when a terminal measurement producing them is part of the experiment. Tests include determining report events and the continuation interface: legal actions, remaining budget, and every clock value affecting future experiments. They are closed under legal one-step continuation, finite randomization, and the bounded monotone limits needed to determine the designated laws.

For an admissible prepared belief b , let $\mathbf{e}_t(b)$ be the evaluations of a countable determining family of these tests, together with the continuation interface. Equality of $\mathbf{e}_t$ is *predictive equivalence*. The family determines the entire designated continuation law, not merely the finite scoring menu introduced below. Here and throughout, a statement about an unavailable action is not a continuation assertion.

We impose the following realization certificate, verified from the instruments. The admissible belief domain $\mathcal{B}_t$ and the action domain are compact metric. The determining evaluations are continuous on $\mathcal{B}_t$. For a continuous report component there is a compact metric report domain Y , a finite dominating measure λ with full support on Y , and an instrument density $T_{t,a,y}$. For every determining next-time test f , the functions

$$(2) \quad n_f(b, a, y) = bT_{t,a,y}f, \quad g(b, a, y) = bT_{t,a,y}1$$

are jointly continuous. The functions $b \mapsto \int_E n_f(b, a, y)\lambda(dy)$, for report events E in a determining algebra, are evaluations of executable one-step continuations, hence are constant on predictive fibres. The same is required for $f = 1$. This is an *event-instrument closure*, not the assertion that a report singleton has positive probability. The normalized belief update $U_{t,a,y}$ preserves the declared domain and is jointly continuous where $g > 0$. If it is used at $g = 0$, a continuous, predictive-fibre-compatible extension is supplied separately. Finite report components use the corresponding atomic instruments. Deterministic reveal components may instead be handled by an explicitly verified continuous quotient update, as in Section 7.2.

Lemma 1.1 (Measurable factorization with completed histories). *Let $R : \Omega \rightarrow E$ and $S : \Omega \rightarrow S_0$ be random variables, where E is standard Borel and S_0 is a nonempty Borel subset of a countable product of compact intervals. If every coordinate of S is measurable with respect to the completion of $\sigma(R)$ under P , there is a Borel map $f : E \rightarrow S_0$ such that $S = f(R)$ almost surely. It is unique P_R -almost everywhere among factors of S .*

Proof. A completed-measurable real random variable has an uncompleted-measurable version: replace the finitely many measurable sets in each dyadic simple approximation by $\sigma(R)$ -sets differing only by null sets, and take a pointwise limit on the common full-measure set. Every set in $\sigma(R)$ is $R^{-1}(B)$ for some Borel $B \subset E$, since these inverse images already form a sigma algebra. The same simple approximations therefore give Borel coordinate factors on E ; assign zero wherever their limits fail to exist. Combine the countably many factors into a product-valued Borel map f_0 . Since S_0 is Borel, $\{e : f_0(e) \in S_0\}$ is Borel and has full P_R -measure. Replace its complement by a fixed point of S_0 . This proves existence. Equality of two factors after composition with R is exactly their equality P_R -almost everywhere. $\square$

Proposition 1.2 (Predictive realization and pointwise update descent). *Under the preceding certificate, $S_t = \mathbf{e}_t(\mathcal{B}_t)$ is compact metrizable and realizes predictive equivalence. Report laws descend continuously. The normalized update descends continuously on the positive-density domain; a declared continuous fibre-compatible extension descends on its declared extended domain. In particular, equivalent beliefs have equivalent updated beliefs for every positive-density report, not just almost every report. Any standard-Borel-valued history statistic through which the determining evaluations and continuation interface are completed-measurable admits a measurable predictive factor almost surely. Setwise sufficiency gives a unique setwise factor on the image, but no additional Borel conclusion for an arbitrary measurable quotient.*

Proof. Rescale the determining evaluations to $[-1, 1]$. Their continuous map into a countable product has compact metrizable image. The determining property identifies its fibres with equality of the designated continuation laws.

Fix equivalent b, b' and an available action a . Event-instrument closure gives equal integrals of $n_f(b, a, \cdot)$ and $n_f(b', a, \cdot)$ over a determining algebra. Uniqueness of finite signed measures extends this equality to all Borel events, and equality of densities follows λ -almost everywhere. A continuous function vanishing almost everywhere under a full-support measure vanishes everywhere: any nonzero value would have a same-sign neighborhood of positive measure. Hence the two numerator functions, and the two denominators, agree for every y . Where $g > 0$, dividing proves equality of every determining next-time evaluation. The continuation interface is among these evaluations. At a zero denominator only the separately verified extension is used.

A continuous surjection from a compact space onto a Hausdorff space is a quotient map. Its product with a compact action or report domain has the same property. Thus the continuous fibre-constant report evaluations and extended updates descend. Restriction to the saturated open positive-density domain also descends, since that domain is the inverse image of an open set under the descended continuous g . Apply Lemma 1.1 to the countable predictive coordinates for measurable minimality. Setwise minimality is the elementary inclusion of sufficient-statistic fibres in predictive fibres. $\square$

This proposition fixes conditional versions rather than inferring them from null events. Finite exact interface components may be included as a disjoint compact union. A noncompact report or belief domain needs a separate Borel realization or a verified localization. No universal smooth realization of an arbitrary predictive equivalence relation is asserted.

1.3. The common prediction task. At time N , after the label has been retained, an independent index $i \in \{1, \dots, q\}$ is selected with fixed positive weights v_i , $\sum_i v_i = 1$. The corresponding legal terminal probe returns $T_i \in [0, 1]$. The algorithm reports $a_i \in [0, 1]$ and incurs $(T_i - a_i)^2$. Each

expectation refers to its own executable probe; a joint counterfactual vector of all probe outcomes is unnecessary. For each possible terminal time t , write $T_{t,i}$ for its corresponding executable probe; at the selected horizon abbreviate $T_i = T_{N,i}$. Put

$$(3) \quad p_t(s) = (\sqrt{v_i} \mathbb{E}[T_{t,i} | s])_{i=1}^q, \quad B_N = \sum_i v_i \mathbb{E} \text{Var}(T_i | H_N), \quad \nu_N = \text{law}(p_N(s_N)).$$

A varying terminal time uses the corresponding declared probe. The finite scoring menu need not determine the entire predictive quotient. Our assumptions below require only its Lipschitz upper bound globally and a separating lower bound on each acquired chart. Unmeasured interface coordinates cannot be silently charged to this task's lower bound.

A checkpoint encoder reads H_N and retains at most m labels. An online encoder has at most m persistent labels at every time, updates from its previous label and the current input/report, and has no readable history tape. A terminal decoder may depend on the public horizon and on the freshly selected probe index, but not on discarded observations. Shared randomness does not carry observation-dependent state. The risks $R_{\text{cp}}(N, m)$ and $R_{\text{on}}(N, m)$ are infima of the full expected score over these respective classes, under the fixed exploration β . Temporary workspace, numerical input access, and program description are recorded separately in Section 5.

The mathematical label class allows Borel transitions on exact current reports. Its risk is not the risk of a finite-bit processor. For an implementation with finite input precision and arithmetic, we write R_{alg} for its actual risk and retain the errors u_t and v_N^{num} explicitly. The latter is a deterministic upper bound on the terminal score-norm error; a stated L^2 bound could be used instead. Appendix A supplies both types of algorithms. A fixed public clock may give the predetermined horizon or protocol phase, but cannot encode observation-dependent past information through adaptive timing. Any such timing channel is retained state and is charged.

2. THE ACQUIRED-GEOMETRY TRANSFER THEOREM

We state one central class theorem, with an explicit kernel-derived route to its causal hypothesis. Its assumptions consist of experimental certificates, not an assumed optimal quantization curve. Throughout, J and d_* are finite and $D_0 > 0$. The geometric constants c, C depend only on $J, d_*, b, B, C_g, L, C_0, D_0$ and, in the separated case, the dimensionless separation ratios D_0/Δ and LD_0/Δ . They do not depend on N, m , the positive widths or acquired weights. Transport enters separately through the displayed Γ_N . For a parameterized family, the following certificates, programs and local error bounds must hold uniformly over the declared family; parameters unknown to the machine cannot occur in its instructions.

2.1. Geometric certificates extracted from histories. Let (D, d) be a compact valid predictive-state domain, with $\text{diam } D \leq D_0$, containing the states under consideration and all representatives used by the algorithm. This is a *valid envelope*, not a claim about the support of the acquired law. Updates on D must have the declared physical meaning or a specified conditional version. Observation-dependent interface components are included and charged. For a fixed exogenous protocol clock, the state fibres at successive times may be identified with D ; the clock remains a declared public argument of the updates, not an observation-dependent memory channel. Suppose the task map satisfies

$$(4) \quad \|p_N(s) - p_N(s')\| \leq Ld(s, s').$$

For each N , disjoint measurable events $E_{N,j}$ partition the actual history law up to a null set. Write $w_{N,j} = P^\beta(E_{N,j})$. Zero-weight components can be omitted. The following conditions are checked by integrating the raw preparation and kernels, not by choosing a convenient auxiliary geometric measure.

(G) Actual mass and global geometry. There are valid compact component domains $D_j \subset D$ covering D , with dimensions $0 \leq d_j \leq d_*$, and maps

$$\phi_j : Q_j \longrightarrow D_j, \quad Q_j = \prod_{k=1}^{d_j} [-a_{j,k}/2, a_{j,k}/2], \quad a_{j,1} \geq \cdots \geq a_{j,d_j} > 0.$$

We normalize the coordinates so that $a_{j,1} \leq D_0$. Zero widths are removed, not inverted. Dimension zero means a singleton. The maps are onto, and, for constants $b, B > 0$,

$$(5) \quad b\|x - x'\| \leq \|p_N(\phi_j x) - p_N(\phi_j x')\| \leq Ld(\phi_j x, \phi_j x') \leq LB\|x - x'\|.$$

Conditioned on $E_{N,j}$, the state law is $\phi_{j\#}(f_{N,j} dx)$, where

$$(6) \quad 0 \leq f_{N,j}(x) \leq C_g/|Q_j|, \quad \int_{Q_j} f_{N,j}(x) dx = 1.$$

For dimension zero this means unit mass. In particular these are actual acquired masses; f need not be positive everywhere in the envelope. For $J > 1$, the compact score images $p_N(D_j)$ have pairwise distance at least $\Delta > 0$. Constants below may depend on D_0/Δ and LD_0/Δ . The components can be curved; no differentiable structure, covariance, or Fisher matrix is prescribed. The same component envelope and grids must serve all intermediate updates of the chosen implementation. A certificate depending on N is permissible, but a uniform-in- N conclusion requires uniform constants and one stated implementation for each budget.

Set $\psi_j(0) = D_0^2$ and, for integers $k \geq 1$, define

$$(7) \quad \psi_j(k) = \begin{cases} \max_{1 \leq l \leq d_j} \left(\frac{\prod_{i=1}^l a_{j,i}}{k} \right)^{2/l}, & d_j > 0, \\ 0, & d_j = 0. \end{cases} \quad \Psi_N(m) = \min_{\substack{k_j \in \mathbb{N} \cup \{0\} \\ 1 \leq \sum_j k_j \leq m}} \sum_j w_{N,j} \psi_j(k_j).$$

There are finitely many allocations, so the minimum is attained. The zero-allocation penalty retains the cost of discarding an entire acquired component. In the case $J = 1$ it is never needed. Formula (7) is explicit in widths and probabilities obtained from the experiment; it is not a variational definition of optimal distortion.

All objects in (G) may depend on the selected horizon: more fully they are

$$D^{(N)}, \quad D_j^{(N)}, \quad \phi_j^{(N)}, \quad Q_j^{(N)}, \quad a_{j,k}^{(N)}, \quad p_N, \quad d_j^{(N)}.$$

Within a theorem at fixed N we suppress these superscripts. A uniform assertion ranges over exactly those horizons with the displayed constants uniform. For that horizon, the codebook covers every representative state needed at intermediate times, not just the terminal support. Actual intermediate laws may be different. Components of zero acquired weight may be removed only when their transport contributions also vanish. Distinct component labels are counted before duplicate physical representatives are merged; merging never increases the budget.

2.2. Causal certificates. For every allocation (k_j) , finite grids in the domains with $k_j > 0$ and a fixed Borel tie rule define a codebook of at most m representatives. A discarded component is encoded into an existing representative. Write $r_j = \sqrt{\psi_j(k_j)}$. The geometric construction in Lemma 3.2 gives component covering errors at most $C_0 r_j$. The following additional certificate is the causal content.

(I) Innovation-localized implementation. There is a specified parameter-independent update of the representative label, executable from its preceding label, the current command/report, and

the declared finite-precision primitives. Along every admitted common command/report path, the exact and representative states satisfy

$$(8) \quad \begin{aligned} e_0 &\leq C_0 \sum_j B_{0,j} r_j + u_0, \\ e_{t+1} &\leq \gamma_{t+1} e_t + C_0 \sum_j B_{t+1,j} r_j + u_{t+1}, \quad e_t = d(s_t, \hat{s}_t). \end{aligned}$$

The nonnegative measurable coefficients $\gamma_t, B_{t,j}$ are determined by the raw experiment and specified update bounds, independently of the grid allocation and precision budget. They may use latent marks for analysis; the implemented machine must not read these marks. The nonnegative u_t bound calibration and numerical errors in the same metric. One sufficient verification is a same-input Lipschitz bound γ_t for the exact update and a rounding error bounded by $C_0 \sum_j B_{t,j} r_j$. An exact hold or grid-preserving transformation is executed without rounding, and then has $B_{t,j} = 0$. An informative overwrite has $\gamma_t = 0$. These checks concern the actual one-step kernel and the given algorithm, not its optimality.

Define empty products as one and set

$$(9) \quad Z_{N,j} = \sum_{t=0}^N B_{t,j} \prod_{i=t+1}^N \gamma_i, \quad U_N = \sum_{t=0}^N u_t \prod_{i=t+1}^N \gamma_i.$$

(T) Acquisition-weighted transport. For all nonnegative real vectors $z = (z_j)$,

$$(10) \quad \mathbb{E}_{P^\beta} \left(\sum_j Z_{N,j} z_j \right)^2 \leq \Gamma_N \sum_j w_{N,j} z_j^2.$$

This is a moment certificate on a positive transport process specified before choosing the memory budget. For example, the positive semidefinite inequality $\mathbb{E}[Z_N Z_N^T] \preceq \Gamma_N \text{diag}(w_N)$ suffices. The weaker inequality on the nonnegative cone in (10) is all that is used. When a component has zero acquired mass, its transport contribution must vanish almost surely. Verifying this point prevents unacquired geometry from entering an average risk law.

(C) Report continuity and effective primitives. On common continuation interfaces the conditional report kernels obey

$$(11) \quad \|g_t(\cdot \mid s, a) - g_t(\cdot \mid s', a)\|_{\text{TV}} \leq \kappa_t d(s, s').$$

A calibrated numerical implementation may additionally incur a same-state report error ζ_t . The grid descriptions, update routines, readouts, and finite allocation search in (7) are specified by finite programs with declared calibration/numerical input access. Exact algebraic width and weight data suffice for the finite allocation comparisons. A certified constant-factor allocation is also sufficient, changing only the upper constant; arbitrary comparisons of unspecified real oracles are not assumed effective. A metric covering argument alone does not prove bit computability; this effective certificate is required for the word “implementable.” The finite algorithms in Section 7 supply it explicitly. A terminal readout approximation contributes at most v_N^{num} in Euclidean score norm.

(K) A structural alternative to (T). The local innovation coefficients have the marked raw-kernel factor of Section 4, with bounded forcing, an invariant environment law and a dominating preparation, and satisfy the one-step Lyapunov inequality (21). The seed satisfies the initial moment bounds of Theorem 4.1. Construct H by the positive resolvents (22), and Γ_N^{ker} by (24), with the stated zero-weight proviso. These are kernel computations, not hypotheses on an optimal risk or an unrolled forced process.

Theorem 2.1 (Causal acquired geometry and kernel-derived resource transfer). *Fix the raw experiment, preparation, exploration, task, and horizon N . Under the predictive realization certificate*

and (G), (I), (C), and either (T) or (K), there are $c, C > 0$ with the dependence specified above such that, for every integer $m \geq 1$, taking $\Gamma_N = \Gamma_N^{\text{ker}}$ under (K),

$$(12) \quad B_N + c\Psi_N(m) \leq R_{\text{cp}}(N, m) \leq R_{\text{on}}(N, m) \leq B_N + \left(C\sqrt{\Gamma_N\Psi_N(m)} + L\|U_N\|_{L^2} + v_N^{\text{num}} \right)^2.$$

For exact comparable allocation data, the online upper is attained as an inequality by the specified finite representative machine with a minimizing allocation. If only an A_{alloc} -approximate allocation is computable, replace C by $C\sqrt{A_{\text{alloc}}}$ in the upper bound. The exact Borel model and the finite-precision program are distinct: when exact Borel primitives and exact calibration are available, use $u_t = v_N^{\text{num}} = 0$; the latter has the displayed nonzero error allowances. If $\Psi_N(m) = 0$, matching means both exact excess risks are zero, not a ratio of zero quantities. In particular, with exact arithmetic and calibration, and $\sup_N \Gamma_N < \infty$,

$$(13) \quad R_{\text{cp}}(N, m) - B_N \asymp R_{\text{on}}(N, m) - B_N \asymp \Psi_N(m)$$

uniformly over the certified horizons, positive widths, and acquisition weights. Widths may vanish and the acquired rank may change, provided the surviving-coordinate certificates have the same constants. There is no inverse of a vanished width in the conclusion.

For a common bounded decision loss of range at most one, the same implementation yields a finite-horizon report-law allowance

$$(14) \quad \min \left\{ 1, \sum_{t < N} \mathbb{E}[\kappa_t e_t + \zeta_t] \right\},$$

whenever the local implementation and report certificates hold for the controller actually compared. An estimate using a moment certificate for a different controller is not implied. Uniform comparison over a controller class requires all prefix certificates used in the bound to be uniform over that class. A common visible stopping time $\sigma \leq N$ inserts $\mathbf{1}_{\{t < \sigma\}}$ in this sum. The assertion is for the given horizon or the stated bounded stopping time, not an infinite-horizon path law. Additional causal simulators compose this allowance with their certified defects, and compose their resource maps as in Theorem 5.1. None of these assertions compares two different full-history oracle baselines.

The acquisition term can be made explicit without inventing a universal power of N . For a specified common ideal baseline b_* with $B_N \geq b_*$, put $I_N = B_N - b_*$. Then the exact implementation has

$$(15) \quad R_{\text{on}}(N, m) - b_* \asymp I_N + \Psi_N(m).$$

The quantity I_N must itself be computed or bounded from the actual acquisition kernel. Section 7.2 computes it exactly as a failed-acquisition probability times a target variance. Equation (15) is not a claim that every experiment has the same statistical exponent.

3. PROOF OF THE TRANSFER THEOREM

Lemma 3.1 (Checkpoint projection). *For the task (3),*

$$(16) \quad R_{\text{cp}}(N, m) = B_N + \inf_{c_1, \dots, c_m \in \mathcal{C}} \int \min_l \|x - c_l\|^2 \nu_N(dx), \quad \mathcal{C} = \prod_i [0, \sqrt{v_i}].$$

Randomized encoders and decoders do not improve this infimum.

Proof. For each executed probe, condition on H_N and on any independent encoder randomization. The cross term in

$$(T_i - a_i)^2 = (T_i - \mathbb{E}[T_i | H_N])^2 + (\mathbb{E}[T_i | H_N] - a_i)^2 + 2(T_i - \mathbb{E}[T_i | H_N])(\mathbb{E}[T_i | H_N] - a_i)$$

has conditional expectation zero. Sum with weights v_i . For a retained label, replace a randomized decoder by its conditional mean; convexity cannot increase the risk. For fixed centers, replace a

randomized label by a nearest center with the first-index tie rule. This is measurable. Conversely, this rule is a legal checkpoint encoder and attains its codebook distortion. Independent shared random seeds only average the costs of such deterministic codebooks, so cannot lower their infimum. Projection of a center to the box $\mathcal{C}$ is nonexpansive relative to all points of $\mathcal{C}$. Compactness gives attainment of the codebook minimum, although only its infimum is needed. $\square$

Lemma 3.2 (Actual anisotropic mass and allocation). *Under (G), the distortion in (16) lies between $c\Psi_N(m)$ and $C\Psi_N(m)$ for every $m \geq 1$. Each component with $k \geq 1$ has a k -point valid-state cover of radius at most $B\sqrt{d_*\psi_j(k)}$. The conclusions hold with the same constants when any collection of widths is removed at zero.*

Proof. Fix a component of positive dimension d and write $h = \sqrt{\psi_j(k)}$. Put $R = 2h$. For coordinates with $a_i > R$, divide the interval into $n_i = \lceil a_i/R \rceil$ equal pieces; for the other coordinates take one piece. If there are l active coordinates, they are the first l , and

$$\prod_i n_i \leq \prod_{i=1}^l \frac{2a_i}{R} = \frac{\prod_{i=1}^l a_i}{h^l} \leq k.$$

The assertion also holds when $l = 0$. Every coordinate interval has length at most R . Its midpoint product grid has Euclidean covering radius at most $\sqrt{d}h$, and (5) gives the valid-state radius. A point component takes one label and has zero radius. An omitted component can use any retained representative with error at most D_0 . Minimizing the allocation proves the upper bound, including the cases $m < J$ and atoms of arbitrary weight.

For the lower bound, a score ball of radius r has preimage of Euclidean diameter at most $2r/b$. If it meets the component, its projection on any one coordinate is contained in an interval of length at most $4r/b$. Integrating the actual density in (6) over the remaining coordinates gives, for $1 \leq l \leq d$,

$$(17) \quad P^\beta\{p_N(s_N) \in B(c, r) \mid E_{N,j}\} \leq C_g \frac{(4r/b)^l}{\prod_{i=1}^l a_i}.$$

The center c need not lie on the image. This is a bound on the actual experiment, not on a fictitious uniform distribution.

If $J = 1$, all m centers can be used in the union bound. For a fixed l , choose $r = c_l(\prod_{i=1}^l a_i/m)^{1/l}$ so that the union has probability at most $1/2$. The expected squared error is at least $r^2/2$. Maximize over l .

If $J > 1$, assign each decoder center to a component whose score image is at distance less than $\Delta/3$, if such a component exists. Separation makes this assignment unique. Let k_j be the number assigned. For $r < \Delta/3$, unassigned or differently assigned centers do not cover any point of component j . When $k_j \geq 1$, use (17) and the preceding radius, capped at $\Delta/4$. Since $\psi_j(k_j) \leq D_0^2$, the cap changes the lower-bound constant only by a factor depending on D_0/Δ , b , and C_g . For $k_j = 0$, the component's distortion is at least $\Delta^2/9$, hence at least a fixed multiple of $\psi_j(0)$. At a point with $k_j \geq 1$ the required lower bound is zero. Multiply each component bound by its actual weight and sum. We have $\sum_j k_j \leq m$. If this sum is zero, give one component a label: its ψ_j cannot increase, since $a_{j,1} \leq D_0$. Thus the sum is bounded below by a constant times the minimum in (7). This proves the lower bound for arbitrary decoder centers and all budgets.

Only surviving coordinates were used. No constant in the covering or small-ball argument contains a_i^{-1} except within the displayed physical volume, which is eliminated in the final scale. This proves the rank-uniform assertion. $\square$

Lemma 3.3 (Innovation transport). *Under (I), for every admitted path,*

$$(18) \quad e_N \leq C_0 \sum_j Z_{N,j} r_j + U_N.$$

Consequently (T) implies

$$\|e_N\|_{L^2} \leq C_0 \sqrt{\Gamma_N \sum_j w_{N,j} r_j^2} + \|U_N\|_{L^2}.$$

Proof. Induct in (8), distributing each nonnegative multiplicative coefficient. This gives precisely (9) and (18). Minkowski's inequality and (10), evaluated at $z_j = r_j$, give the second assertion. No independence of the coefficients is used. $\square$

Proof of Theorem 2.1. Under (K), Theorem 4.1, whose proof uses only the raw marked kernel and positive resolvents, first supplies (T). This invocation is not circular. Lemma 3.1 and Lemma 3.2 give the lower bound for every checkpoint encoder. The terminal output of any online encoder is a checkpoint encoder, so $R_{\text{cp}} \leq R_{\text{on}}$. Choose a minimizing allocation in (7) and its valid grids. With only a certified A_{alloc} -approximate allocation, use it instead; its weighted squared radii increase by at most A_{alloc} . Execute the label update in (I). By (4), its score error has L^2 norm at most $L\|e_N\|_{L^2} + v_N^{\text{num}}$. The conditional projection identity is valid for this online encoder as well. Apply Lemma 3.3, and absorb LC_0 into C . This proves (12). Its exact-arithmetic specialization gives (13); adding the same nonnegative I_N yields (15).

For path comparison, run the exact and representative report experiments with the same controller at identical visible prefixes. At such a prefix the action kernels coincide, the retained label is the same function of the prefix, and the next-report kernels differ by at most $\kappa_t e_t + \zeta_t$. Maximal coupling of the next report is possible on standard Borel spaces. The common-prefix subprobability law is dominated by the exact prefix law. Hence the probability of a first disagreement at step $t+1$ is at most $\mathbb{E}_{P^\beta}[\kappa_t e_t + \zeta_t]$, or the corresponding expectation for the controller actually being compared. Summing proves (14). Append the common label update and resource readout; these do not increase the discrepancy. After the common stopping decision, both processes are absorbed, which inserts the active-step indicator. To use the numerical bound from (T) for a different controller, that controller must have its own certificate; the fixed- β certificate is not automatically uniform over policies. Composition and common-loss transfer are proved in Section 5. $\square$

3.1. How to verify the transport certificate.

Proposition 3.4 (Three kernel-level mechanisms). *The following are sufficient verifications of (T).*

(a) *With one component of weight one, $\gamma_t \leq \rho < 1$, $B_{t,1} \leq 1$, and $B_{0,1} \leq 1$, one may take $\Gamma_N \leq (1 - \rho)^{-2}$.*

(b) *Suppose each update is either an exact hold with $\gamma_t = 1$, $B_{t,1} = 0$, or an active step with $\gamma_t \leq \rho < 1$, $B_{t,1} \leq 1$. Then the same constant works, without a lower bound on the probability or frequency of active steps. Here $w_{N,1} = 1$; an initial rounding is permitted.*

(c) *Suppose the process starts at one known unresolved point, waits there, and at a first successful acquisition moves into component j , thereafter holding. The update on a successful report overwrites the old prediction, including its conditional version at an otherwise impossible report. Let $B_{0,0} = 1$, and let $B_{t,j} = 1$ only at the acquisition into j . Then*

$$Z_{N,0} = \mathbf{1}_{\{\text{no acquisition by } N\}}, \quad Z_{N,j} = \mathbf{1}_{\{\text{acquired component } j \text{ by } N\}}.$$

For the corresponding actual component weights, (10) is an equality with $\Gamma_N = 1$.

Proof. In (a), sum a geometric series. In (b), delete the exact holds from the error recurrence, not from the raw acquisition or time account. The resulting pathwise series is the same geometric series indexed by active events. In (c), failure and post-acquisition holds have $A = 1, B = 0$, while the acquisition has $A = 0$ and the appropriate $B = 1$. Thus the initial contribution is killed by the first acquisition and the new contribution persists. The displayed indicators are disjoint and exhaustive. Squaring their weighted sum and taking expectation gives the assertion. $\square$

Mechanism (b) permits arbitrarily long metastable waiting periods. It is different from replacing the raw horizon by an uncharged number of successful reports. Mechanism (c) additionally preserves the small acquired mass in the upper bound, rather than paying the full geometric cost on paths that never acquire it. Recurrent nonuniform stability is derived from the one-step kernel condition in Theorem 4.1; it is not inferred from a negative average logarithmic Lipschitz exponent alone. The relation to average contraction theory [2] is thus precise: here the required object is a forced, acquisition-weighted second moment, not merely convergence of unforced iterates.

Corollary 3.5 (Intersecting strata). *Remove the separation assumption from (G), allow the score images of the J component charts to intersect, and let the raw preparation carry a latent component mark with probabilities $w_{N,j}$. The mark need not be available to the algorithm or retained by the quotient. Suppose the conditional laws and the other certificates remain valid, with chart rounding certified conditionally on the raw mark. Then (12) and (13) hold for every $m \geq 2J$, with constants also depending on J , and without any inverse distance between strata.*

Proof. The global upper cover is still the union of the component grids. Encode into a nearest valid representative in d ; its error on component j is no greater than that component grid's error. Duplicate representatives may be merged. The algorithm does not need the latent mark.

For the lower bound apply (17) with all m decoder centers to each positive-dimensional conditional law. This gives distortion at least $c \sum_j w_{N,j} \psi_j(m)$; point components contribute zero. Set $k = \lfloor m/J \rfloor \geq 1$. Directly from (7), for $m \geq k \geq 1$,

$$\psi_j(k) \leq (m/k)^2 \psi_j(m) \leq (2J)^2 \psi_j(m).$$

The allocation of k labels to every component is feasible. Thus $\Psi_N(m) \leq (2J)^2 \sum_j w_{N,j} \psi_j(m)$. Combining the last two bounds proves the claimed lower, and the remaining causal proof is unchanged. In particular neither smoothness at an intersection nor identifiability of the latent mark was used. $\square$

4. A KERNEL THEOREM FOR RECURRENT NONUNIFORM TRANSPORT

The moment condition in the transfer theorem is useful only insofar as it can be derived without solving the approximation problem itself. We now derive it from a positive operator attached to a marked factor of the raw experiment. The factor may be used only for analysis. It need not be observed, and it is never supplied to the encoder.

4.1. One-step data and positive resolvents. Let Ξ be standard Borel, and suppose that the coefficients of the innovation recurrence are generated by a time-homogeneous marked kernel

$$(19) \quad P(d\xi', d\gamma, d\mathbf{b} \mid \xi), \quad \gamma \geq 0, \quad \mathbf{b} \in [0, \infty)^J, \quad \sum_j b_j \leq B_*.$$

Thus, conditionally on ξ_t , the next environment and marks are independent of all previous marks and of the seed. This assertion is checked from the raw preparation, the chosen exploration and the one-step update bounds. It excludes coefficients chosen after seeing the grid allocation. Controller state may have to be included to make the factor Markov; its implementation cost is not thereby removed.

Assume that the environment marginal has an invariant probability μ . Disintegrate the stationary marked edge measure in the reverse direction:

$$\mu(d\xi) P(d\xi', d\gamma, d\mathbf{b} \mid \xi) = \mu(d\xi') \overleftarrow{P}(d\xi, d\gamma, d\mathbf{b} \mid \xi').$$

All the following identities are statements about measurable versions μ -almost everywhere. Define, for nonnegative measurable f ,

$$(20) \quad \begin{aligned} (K_p f)(\xi') &= \int \gamma^p f(\xi) d\overleftarrow{P}, & p &= 1, 2, \\ f_j(\xi') &= \int b_j d\overleftarrow{P}, & g_{jk}(\xi') &= \int b_j b_k d\overleftarrow{P}, \\ (D_j f)(\xi') &= \int \gamma b_j f(\xi) d\overleftarrow{P}. \end{aligned}$$

The structural hypothesis is the one-step inequality

$$(21) \quad 1 \leq v \leq v_+ < \infty, \quad K_2 v \leq \lambda v, \quad 0 \leq \lambda < 1.$$

Here v is a measurable Lyapunov weight. Neither $\gamma < 1$ on each edge nor a bound on a previously unrolled error is assumed. The operators depend only on the experimental factor and the selected local update/rounding bounds.

For a positive weight u , write $\|f\|_u = \text{ess sup } |f|/u$. Cauchy–Schwarz gives $K_1 \sqrt{v} \leq \sqrt{\lambda} \sqrt{v}$. Hence the positive Neumann resolvents $\mathcal{S}_1 = (I - K_1)^{-1}$ on the $\sqrt{v}$ -weighted space and $\mathcal{S}_2 = (I - K_2)^{-1}$ on the v -weighted space exist. Set

$$(22) \quad \begin{aligned} h_j &= \mathcal{S}_1 f_j, \\ V_{jk} &= \mathcal{S}_2 (g_{jk} + D_j h_k + D_k h_j), & H_{jk} &= \int V_{jk} d\mu. \end{aligned}$$

The profiles are finite. Indeed h_j is bounded by a constant times $\sqrt{v}$, and Cauchy–Schwarz bounds $D_j h_k$ by a constant times $\sqrt{v}$, hence by a constant times v . The constants depend only on $B_*, v_+, 1 - \lambda$ and J , not on a memory allocation.

Theorem 4.1 (Forced moment resolvent from a raw-kernel factor). *Suppose (19)–(21) hold. Let*

$$Z_{0,j} = 0, \quad Z_{t+1,j} = \gamma_{t+1} Z_{t,j} + b_{t+1,j}.$$

Under the stationary environment preparation, for every $N \geq 0$ and every $z \in [0, \infty)^J$,

$$(23) \quad \mathbb{E} \left(\sum_j z_j Z_{N,j} \right)^2 \leq z^\top H z.$$

The matrix H is symmetric positive semidefinite and has nonnegative entries. The same inequality holds with a nonnegative seed if, conditionally on ξ_0 , its first moments are bounded by h_j and its mixed second moments by V_{jk} , entry by entry. If the actual initial environment law has density at most D_ with respect to μ , and uses the same conditional seed rule, the right side is multiplied by D_* .*

Consequently, for the actual acquired weights $w_{N,j}$, condition (T) follows with

$$(24) \quad \Gamma_N^{\text{ker}} = D_* \left\| \text{diag}(w_N)^{-1/2} H \text{diag}(w_N)^{-1/2} \right\|_{\text{op}},$$

provided every zero-weight row of H is zero; the inverse is taken only on positive-weight coordinates. If this proviso fails, this stationary majorant does not certify that horizon. A finite-horizon profile or another mechanism must then be used. In particular, for $J = 1$, $w_{N,1} = 1$, the one-step drift alone gives a uniform finite transport constant, with no assumption of uniform contraction of the active updates.

Proof. First start the environment at μ and the forcing process at zero. Define conditional moment profiles

$$m_{t,j}(\xi) = \mathbb{E}[Z_{t,j} \mid \xi_t = \xi], \quad C_{t,jk}(\xi) = \mathbb{E}[Z_{t,j} Z_{t,k} \mid \xi_t = \xi].$$

The Markov-factor hypothesis ensures conditional independence of future edge marks and the previous forcing history given the previous environment. Conditioning on the stationary incoming edge therefore gives the exact recursions

$$(25) \quad \begin{aligned} m_{t+1,j} &= K_1 m_{t,j} + f_j, \\ C_{t+1,jk} &= K_2 C_{t,jk} + D_j m_{t,k} + D_k m_{t,j} + g_{jk}. \end{aligned}$$

Truncation of the marks justifies these identities before integrability is known; monotone convergence then removes the truncation. Alternatively, induction using the finite bounds below proves integrability at each finite time.

Positivity and the identities $h_j = K_1 h_j + f_j$ and $V_{jk} = K_2 V_{jk} + D_j h_k + D_k h_j + g_{jk}$ imply $0 \leq m_{t,j} \leq h_j$ and $0 \leq C_{t,jk} \leq V_{jk}$ by induction. In fact m_t and C_t increase from zero. The first recursion converges in the $\sqrt{v}$ norm to h . Subtracting the second recursion from its fixed-point identity, or summing its geometric convolution, proves convergence in the v norm to V . To see the latter directly, split the convolution at a fixed lag: its long-lag part is bounded by a geometric λ -tail, and the finitely many remaining terms converge because $m_t \rightarrow h$.

For each t the conditional matrix $C_t(\xi)$ is positive semidefinite. Entrywise convergence in a finite matrix preserves this property outside one common null set. Thus $V(\xi)$ and its integral H are positive semidefinite. All entries are nonnegative. Integrating the entrywise upper bounds and multiplying by $z_j z_k \geq 0$ proves (23). This last nonnegative-cone bound is the one needed for covering radii; it is not an entrywise-to-Loewner inference.

For a seed satisfying the stated conditional inequalities, the same induction in (25) applies, without the monotone-convergence assertion for that seed. An initial density bounded by D_* dominates the entire path-and-seed law by D_* , since all subsequent kernels agree. Integrating any nonnegative path functional proves the domination claim. Finally apply the operator norm bound to $\text{diag}(w_N)^{1/2} z$. A zero diagonal of a positive semidefinite matrix forces its row to be zero; the explicit proviso prevents assigning positive transport energy to a component of zero acquired probability. $\square$

This theorem replaces a path-length family of forced-moment inequalities by one-step kernel integration, a Lyapunov inequality, and two positive resolvents. The comparison with actual acquired weights remains visible in (24): small mass is not made free by a stability theorem. Uniform matching across a family means that this explicitly computed normalized matrix is uniformly bounded. For one acquired chart no further mass-normalization condition is needed. For a finite environmental chain, all these objects are finite matrices and vectors, independent of N and m .

Corollary 4.2 (Finite-state check and recurrent expansion). *For a finite environment, $\rho(K_2) < 1$ is equivalent to the existence of a strictly positive vector v and $\lambda < 1$ satisfying $K_2 v \leq \lambda v$. In particular consider a two-state chain, with states C, E , transition matrix*

$$P = \begin{pmatrix} 1-p & p \\ 1 & 0 \end{pmatrix}, \quad 0 < p \leq 1,$$

and gain $1/4$ on arrival at C , 2 on arrival at E . Under its invariant law, the reversed chain has the same transition matrix and

$$K_2 = \begin{pmatrix} (1-p)/16 & p/16 \\ 4 & 0 \end{pmatrix}, \quad K_2 \begin{pmatrix} 1 \\ 8 \end{pmatrix} \leq \frac{1}{2} \begin{pmatrix} 1 \\ 8 \end{pmatrix}.$$

Bounded forcing therefore satisfies Theorem 4.1, although expansions recur and the conditional squared gain on arrival at E is four.

Proof. A positive vector inequality makes the weighted sup norm of K_2 at most λ , so its spectral radius is less than one. Conversely, when $\rho(K_2) < 1$, let $v = (I - K_2)^{-1} \mathbf{1} \geq \mathbf{1}$. Then $K_2 v = v - \mathbf{1} \leq (1 - 1/\max_i v_i) v$; the zero matrix case is included. The displayed chain has invariant probabilities $(1, p)/(1 + p)$ and satisfies detailed balance. Multiplication by the squared incoming

gain gives the stated matrix. Its first row applied to $(1, 8)$ is $(1 + 7p)/16 \leq 1/2$ and its second is four. These calculations prove the example. It is an operator witness; a complete raw experiment and its acquired geometry are verified separately below. $\square$

Proposition 4.3 (Exact suspension preserves the forced profiles). *Replace each transition of (19) independently by an exact hold with probability $1 - \theta$, $0 < \theta \leq 1$. A hold has unchanged environment, gain one and zero forcing. Keep every hold in the raw-call and physical-time accounts. The invariant environment law remains μ , and the profiles h, V, H in (22) are unchanged. Thus the bound (24) has no inverse power of θ apart from any change in the independently computed acquired weights. At $\theta = 0$ the forcing process remains its seed, so the same upper bound holds for the seeds admitted in Theorem 4.1.*

Proof. Reversing the stationary held kernel gives the same mixture. Therefore

$$K_p^\theta = (1 - \theta)I + \theta K_p, \quad f_j^\theta = \theta f_j, \quad g_{jk}^\theta = \theta g_{jk}, \quad D_j^\theta = \theta D_j.$$

The drift constant becomes $1 - \theta(1 - \lambda) < 1$. Cancelling θ in the two fixed-point equations proves $h^\theta = h$ and $V^\theta = V$. This cancellation is justified by the positive resolvent's uniqueness, not by replacing physical time with a random stopping time. At zero activity there are no new innovations and direct retention proves the assertion. The independence and state-independent value of θ are part of this proposition; arbitrary adaptive suspensions require their own kernel calculation. $\square$

For a scalar independent active gain G , with forcing one at every active step, write $a_1 = \mathbb{E}G$ and $a_2 = \mathbb{E}G^2 < 1$. The equations give

$$(26) \quad h = \frac{1}{1 - a_1}, \quad H = \frac{1 + a_1}{(1 - a_1)(1 - a_2)}.$$

Indeed $a_1 \leq \sqrt{a_2} < 1$, and substitution into $H = a_2 H + 2a_1 h + 1$ proves the formula. It also bounds any seed with first moment at most h and second moment at most H . This explicit second-moment formula is a consequence of the general kernel theorem, not a replacement for it. Section 7.3 verifies a recurrent continuous-state experiment in this class, including its actual density and finite program.

A negative average logarithmic gain would not replace (21). For example, $G = 1/4$ with probability $9/10$ and $G = 4$ with probability $1/10$ has negative mean logarithm but $\mathbb{E}G^2 = 53/32 > 1$. Already the squared product multiplying one earlier nonzero innovation has expectation growing geometrically. The theorem controls forced second moments, not only almost-sure contraction of unforced iterates.

5. CAUSAL MORPHISMS AND THE FULL RESOURCE ACCOUNT

We now make explicit the implementation structure through which a geometric risk theorem is transferred. In particular, an internal register is not made free by omitting it from a risk argument, and a register appearing in one construction is not automatically mandatory in every construction.

5.1. Algorithms, resources and admissible interfaces. Fix a parameter set Λ , raw experiments $\mathfrak{E}_\lambda$, a common task with loss in $[0, 1]$, and a class of legal controllers. A controller has no access to λ . A resource-bounded algorithm consists of Borel state-transition kernels, a terminal readout, and, in the finite-bit model, a fixed finite program and a specified numerical input interface. There is no readable discarded-report tape. A random tape is independent of the experiment and immutable; an observation-dependent pointer into it is retained state.

We use the implementation ledger

$$(27) \quad b = (N_{\text{raw}}, m, R, D_c, Q, W, P, b_{\text{cal}}, b_{\text{num}}, \tau).$$

Here N_{raw} counts all raw calls, including pilots, failures and every actually executed terminal probe; m, R, D_c, Q are, respectively, representative, simulator, controller and internal-clock state capacities; W is transient workspace in bits; P is program and readonly-data length in bits; the two precision capacities specify the calibration and numerical input interfaces; arithmetic guard bits are charged

in workspace; and τ is physical elapsed time. Actual numerical error is a separate property of those interfaces and routines, not automatically $2^{-b_{\text{num}}}$. Workspace is reset between calls. Any observation-dependent information surviving a reset belongs to the persistent account. A table and a generator program have different P and time accounts.

The total persistent capacity of a direct product implementation is $M = mRD_cQ$. To discuss arbitrary fused implementations, we also use

$$(28) \quad \mathbf{b} = (N_{\text{raw}}, M, W, P, b_{\text{cal}}, b_{\text{num}}, \tau).$$

A fused machine is constrained by its total number of observation-dependent states, not by a preferred factorization. Both vectors remain explicit in the relevant risk notation. In Sections 1–4, N denotes calls before the forecast; embedding a protocol with one further physical terminal probe requires $N_{\text{raw}} \geq N + 1$. A menu of counterfactual probes is never charged as though they had all been executed. The resource preorder is coordinatewise comparison of capacities; known interface precision requirements are compared in the same direction. Different input interfaces are different algorithm models, even at the same integer capacity.

The superscript χ in $\mathcal{A}_{\mathfrak{E}}^{\chi}(b)$ and $\mathcal{R}_{\mathfrak{E}}^{\chi}(b)$ specifies the cost semantics. Unless indicated otherwise, $\chi = \text{pw}$ means pathwise call and time bounds. For $\chi = \text{mean}$ these two bounds hold in expectation uniformly over the declared parameter/controller class. A tail specification includes its failure probabilities explicitly. Persistent capacities, workspace and readonly-description bounds are pathwise in all three semantics. For example, the parameter-uniform risk is

$$\mathcal{R}_{\mathfrak{E}}^{\chi}(b) = \inf_{A \in \mathcal{A}_{\mathfrak{E}}^{\chi}(b)} \sup_{\lambda \in \Lambda} \mathbb{E}_{\mathfrak{E}_{\lambda}}^A \ell.$$

The fixed-preparation version replaces the supremum by the specified prepared law. The empty-class value is $+\infty$. For the fixed-exploration prediction theorem the algorithm chooses the encoder, not the exploration.

A public clock is free only when it is the fixed exogenous protocol clock, independent of observed values. An adaptively delayed emission, a data-dependent stopping phase that will be read later, or a remembered failure count is not such a clock. Its observation-dependent states enter Q , or enter the total capacity M directly.

5.2. Complete causal-morphism data. A morphism $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ has the following data. Source and target have specified visible filtrations and legal action correspondences. The simulator has an initial state kernel, independent of the unknown parameter, and a microstep kernel. Given its current state, the current target command and the source prefix, this kernel either chooses a legal source action or emits a target report and a new simulator state. Successive target emissions occur at nondecreasing source stopping indices σ_k ; the declared finite call bound prevents infinite internal execution. A terminal command is mapped to a legal source continuation producing the common loss-relevant terminal readout. A target stopping decision is mapped to source stopping, and the published target clocks and costs have specified report transformations. The emitted reports include failures when the target interface includes them. Prefix dependence is a mathematical description, not permission to reread a discarded tape: the resource certificate must implement it through the declared retained state and current inputs.

The resulting target-to-source lift L_{Γ} is part of the data. It maps every admitted target controller into an admitted source controller, preserving action availability, task semantics and the stopping convention. This is a statement to be checked on the declared classes, not merely a type name. The state update and report transformation are measurable on standard Borel domains. All auxiliary randomness is parameter independent. In finite-bit assertions the same data have the executable routines and numerical interfaces charged in (27).

The complete interactive defect is

$$(29) \quad \epsilon_\Gamma(b) = \sup_{\lambda, \alpha \in \mathcal{A}_{\mathfrak{F}}(b)} \left\| \text{law}^{\mathfrak{E}\lambda}(\Gamma; \mathbf{L}_\Gamma \alpha) - \text{law}^{\mathfrak{F}\lambda}(\alpha) \right\|_{\text{TV}}.$$

The compared laws include the target-visible transcript, costs and clocks, the common terminal readout and the controller's terminal decision. They also include any independent controller seed shared with a later readout. Equivalently, it suffices to have a uniform comparison conditional on each such seed. This prevents the unjustified extension of a marginal-transcript bound to a correlated controller decision. Our convention is $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$, one half of the density L^1 distance. Taking the infimum over a declared class of these simulators defines causal deficiency. A particular simulator supplies only an upper certificate.

A resource map ρ_Γ is a monotone map of the full ledgers with a proof that the lifted implementation lies within its image budget. One useful explicit pathwise map is

$$(30) \quad \rho_\Gamma(b) = (n_0 + n_\Gamma N_{\text{raw}}, m, r_\Gamma R, d_\Gamma D_c, q_\Gamma Q, W + w_\Gamma(b), P + p_\Gamma(b), \beta_{\text{cal}, \Gamma}(b), \beta_{\text{num}, \Gamma}(b), \tau_0 + t_\Gamma N_{\text{raw}} + t'_\Gamma \tau).$$

The nonnegative coefficients and monotone functions are known protocol bounds. Pilots are included in n_0, τ_0 . Precision maps specify enough source precision for the target accuracy, not a change of the unknown physical parameter. The workspace sum is a safe simultaneous-storage bound; a proof of reuse may improve it. A direct product machine at the image budget has at most $m(r_\Gamma R)(d_\Gamma D_c)(q_\Gamma Q)$ persistent states. A different verified monotone cost map is permitted; merely writing ρ does not certify one.

Theorem 5.1 (Typed resource and common-risk composition). *Let $\Gamma_1 : \mathfrak{E} \rightarrow \mathfrak{F}$ and $\Gamma_2 : \mathfrak{F} \rightarrow \mathfrak{G}$ have the preceding data, compatible action/readout/clock interfaces, and resource maps ρ_1, ρ_2 . For each admitted outer budget b , assume*

$$\mathbf{L}_{\Gamma_2} \mathcal{A}_{\mathfrak{G}}(b) \subseteq \mathcal{A}_{\mathfrak{F}}(\rho_2(b)), \quad \mathbf{L}_{\Gamma_1} \mathcal{A}_{\mathfrak{F}}(\rho_2(b)) \subseteq \mathcal{A}_{\mathfrak{E}}(\rho_1(\rho_2(b))).$$

Their serial composition is a causal, parameter-independent morphism with resource map $\rho_1 \circ \rho_2$ and defect at most

$$\min\{1, \epsilon_1(\rho_2(b)) + \epsilon_2(b)\}.$$

For a common bounded loss and the same preparation or parameter supremum,

$$(31) \quad \mathcal{R}_{\mathfrak{E}}^{\text{pw}}(\rho_\Gamma(b)) \leq \mathcal{R}_{\mathfrak{F}}^{\text{pw}}(b) + \epsilon_\Gamma(b).$$

For the prediction implementation of Theorem 2.1, with $N_{\text{raw}} \geq N + 1$, keeping every coordinate of b and its image,

$$(32) \quad \mathcal{R}_{\mathfrak{E}}^{\text{pw}}(\rho_\Gamma(b)) \leq B_N + \left(C \sqrt{\Gamma_N \Psi_N(m)} + L \|U_N\|_2 + v_N^{\text{num}} \right)^2 + \epsilon_\Gamma(b) + \Delta_{\text{rep}} + p_{\text{bad}}.$$

Here the fixed-preparation risk is used. With parameter-uniform certificates the parameter-uniform version takes the supremum of the entire right side over the parameter; in particular the Bayes baseline is not assumed parameter independent. The term Δ_{rep} is the bounded-horizon allowance (14) only for a separate report-simulation stage not already counted in ϵ_Γ . Deterministic approximation certificates may fail on an event of probability at most p_{bad} .

Expected-cost versions hold when the inner per-call and pilot bounds hold conditionally at every calling history, uniformly over the lifted class. Tail versions hold with the corresponding joint good-resource event and with the union bound on its declared failure probabilities. Unconditional expected-cost bounds alone do not establish the asserted affine composition.

Proof. Execute the two microstep kernels in their causal order, carrying their joint internal state. Their stopping indices compose because every decision uses only the currently available prefix; the finite resource maps exclude infinite internal execution. The two inclusions in the statement establish

legality of the composed lift. State updates, report readouts, clocks and terminal commands compose as measurable kernels. This constructs the morphism and its full resource map, including precision, workspace and readonly data. In particular independent implementation registers multiply their state capacities, while their ideal logarithmic bit capacities add.

Fix a parameter and an outer controller, including its shared independent seed. Insert between the two path laws the law with an exact intermediate experiment. The first discrepancy is at most $\epsilon_1(\rho_2(b))$ by the lifted-class inclusion and the complete interactive defect. The second nonanticipating transformation is a Markov kernel and cannot increase total variation. The remaining discrepancy is at most $\epsilon_2(b)$. The triangle inequality proves the stated defect. The inclusion of the common terminal readout ensures that this is a comparison of the actual task, not of two surrogate outputs.

For $0 \leq \ell \leq 1$, the layer-cake formula bounds the difference of its expectations by total variation. Apply this to each target algorithm, then take the same parameter supremum and the algorithm infimum. This gives (31), using arbitrarily near-optimal target algorithms when an infimum is not attained. Substitute Theorem 2.1 to obtain (32). On a bad-certificate event the loss is at most one; on its complement the pointwise approximation estimate is integrated against the original law. This argument does not condition the geometric law on the good event.

For mean budgets, write the total inner cost as the pilot cost plus the sum of per-call costs times the indicator that the call is reached. Conditional bounds and the tower property give the affine expectation bound, including random numbers of calls. For tail budgets, intersect the certified good events, compose the pathwise bounds there, and use the union bound for their complement. These arguments also explain the stated restrictions on expected and tail semantics. $\square$

Calibration and arithmetic errors add in the one-step u_t before multiplication by the same transport factors. Minkowski's inequality then adds their propagated root-mean-square contributions before squaring. A simulation defect acts on the common bounded loss and is added linearly. An allowed calibration error or defect need not occur on any particular instance; a converse requires an indistinguishability experiment. Nor does a forward simulator transfer a lower bound in the reverse direction. The next section supplies lower experiments and a mandatory-state theorem rather than making either inference.

6. MANDATORY STATES AND A JOINT MINIMAX EXPERIMENT

The preceding resource map is constructive. Its converse depends on what future continuations must still be executable. We first establish a general state-separation principle, then exhibit a single experiment in which all the proposed risk terms occur simultaneously.

6.1. A continuation-separation cut. A *latched continuation tag* is a variable $C \in \{1, \dots, L\}$, reported before a memory cut and not reissued afterward. A fresh legal challenge after the cut asks for C with zero error. Equivalently, finitely many challenges ask for its coordinates. This is an exact continuation requirement in the algorithm class, not an extra squared-loss penalty with an unstated weight. The prediction loss is unchanged. We call this the *audited continuation interface*. An external clock cannot supply the tag.

Theorem 6.1 (Mandatory-state separation and total-memory resolution). *Let C be uniform on L tags and independent of an acquired score X with law ν . A checkpoint machine has at most M total persistent states and must satisfy the audited continuation interface. Let $Q_\nu(k)$ be its ordinary k -center squared distortion, with arbitrary centers. For $M \geq L$, its exact checkpoint excess risk is*

$$(33) \quad \min_{\substack{k_c \in \mathbb{N}, \\ \sum_c k_c \leq M}} \frac{1}{L} \sum_{c=1}^L Q_\nu(k_c).$$

If $M < L$, the admissible class is empty. These statements include randomized fused implementations with independent shared randomness.

Suppose ν has one anisotropic acquired chart of dimension at most d_* and the constants of (G). Write ψ for its positive-budget scale in (7). Then, for $M \geq 2L$, every such machine has excess at least

$$(34) \quad c\psi(\lfloor M/L \rfloor).$$

If the geometric experiment also satisfies the bounded-transport online theorem and the tag remains unchanged, an online product construction with $L\lfloor M/L \rfloor$ states attains the corresponding upper bound, with its full workspace/program/precision/time costs charged. Constants are independent of M, L and positive widths.

In particular, if a simulator key, a controller continuation token and an internal phase token are independent uniform tags of sizes R_0, D'_0, Q_0 , and the future interface separately challenges them, then $L = R_0 D'_0 Q_0$. The growing product overhead is mandatory even for an implementation that fuses all registers. No assertion of necessity is made for registers lacking these continuation distinctions.

Proof. Fix an independent shared seed. A persistent state reached with positive probability under two different tags cannot support a zero-error answer to the same fresh challenge under both tags. Hence the positive-probability state supports of different tags are disjoint. There are only finitely many tags and states, so one may choose a common full-measure set of seeds on which this holds. Let k_c be the size of the support for tag c . Then $k_c \geq 1$ and $\sum_c k_c \leq M$.

For each tag, the terminal prediction vectors have at most k_c centers. Randomized decoding is replaced by its mean, and randomized encoding cannot improve nearest-center distortion. Independence of C and X therefore gives a conditional excess of at least $Q_\nu(k_c)$. Averaging proves the lower bound in (33). Conversely, use disjoint sets of k_c labels for optimal codebooks under each tag and decode the tag from its set. Compactness gives optimal codebooks in our geometric setting; approximating codebooks suffice in the infimum formulation. This proves equality and infeasibility when $M < L$. Random shared seeds only average costs bounded below by the same minimum.

To prove the uniform scale bound, put $x = M/L \geq 2$. At least half the tags have $k_c \leq 2x$. Lemma 3.2 and monotonicity give an average distortion at least a fixed multiple of $\psi(\lceil 2x \rceil)$. If $k \geq n \geq 1$, directly from (7),

$$\psi(k) \geq (n/k)^2 \psi(n).$$

Since $\lceil 2x \rceil / \lfloor x \rfloor \leq 5$, this proves (34). For the online upper, retain the exact tag and run the certified geometric encoder with $\lfloor M/L \rfloor$ labels. The tag affects neither its acquired law nor its error recurrence. Its product state capacity is at most M . Independent separately challenged tags are distinguished exactly by their joint value, proving the last assertion. $\square$

This is a converse for a semantic continuation requirement, not a proof that each factor in every chosen implementation must be present. With an allowed tag-error probability a different, quantitative state-separation argument would be needed. Likewise a fixed public phase known independently of observations is not one of these tags.

6.2. One common robust experiment. Fix an integer $d \geq 1$, widths $0 \leq a_i \leq 1/4$, an acquisition probability $s \in [0, 1]$, a tag size $L \geq 1$, $0 \leq \delta \leq 1/4$, $0 \leq \varepsilon \leq 1/2$, and an integer $b \geq 1$. Zero-width coordinates below are deterministic. Prepare independently a tag C uniform on L values and a vector

$$X_i \sim \text{Unif}[1/2 - a_i/2, 1/2 + a_i/2].$$

The physical parameter is $\lambda = (\sigma, \eta, z) \in \{-1, 1\}^2 \times \{0, 1\}$. It is not supplied to the algorithm. Put $h_b = 2^{-b-2}$. The common calibration input for the terminal probability $1/2 + \sigma\delta$ is $1/2$; the common b -precision input for the terminal probability $1/2 + \eta h_b$ is also $1/2$. Both points in the latter pair lie in the same width- 2^{-b} rounding cell centered at $1/2$. There is no legal preterminal probe revealing either sign. These are specified information interfaces, not optional inaccurate arithmetic.

The first paid call latches C and supplies these identical calibration/numerical packets. The second reports the bit z with probability $1 - 2\varepsilon$ and an erasure otherwise. The erasure coin is independent of everything else. The next n raw calls are attempts: each unresolved attempt succeeds with independent probability s , reveals X and absorbs; failures and post-success calls are exact holds. Including the one selected terminal probe below, all raw calls count, so $N_{\text{raw}} = n + 3$. The exploration makes these calls, independently of the encoder. At the memory cut the audited continuation interface may challenge C . This challenge is a readout of the retained machine, not an additional physical measurement of the hidden apparatus; its output operations are charged in time.

The soft task independently selects one of four blocks with probability $1/4$. In the geometry block it selects a coordinate uniformly and executes a terminal Bernoulli probe of mean X_i . The other blocks execute a terminal Bernoulli probe of mean $1/2 + \sigma\delta$, a terminal Bernoulli probe of mean $1/2 + \eta h_b$, or the deterministic terminal bit z . Only the selected probe is executed. A forecast in $[0, 1]$ is made before its outcome. Every block uses squared loss, so the common loss lies in $[0, 1]$.

Define the exact-bit target by changing only the second call to an exact report of z , with the same costs, all other kernels and the same terminal menu. Admitted bit simulators randomize the sole preterminal bit report and parameter-independent auxiliary reports. No terminal readout can be requested before the target forecast: the terminal command follows that forecast and ends the physical experiment. Thus a simulator cannot query the future answer to manufacture an earlier exact bit. The deficiency of this source–target pair is exactly ε , by Proposition 6.3; independent auxiliary reports do not change the binary testing lower bound.

The ideal oracle knows X, σ, η, z ; its risk is the same for every λ :

$$(35) \quad b_* = \frac{1}{4} \left\{ \frac{1}{d} \sum_i \mathbb{E}[X_i(1 - X_i)] + (1/4 - \delta^2) + (1/4 - h_b^2) \right\}.$$

This is the single baseline subtracted below. Write

$$H_n = (1 - s)^n, \quad V_a = \frac{1}{12d} \sum_i a_i^2, \quad \psi_a(k) = \max_{1 \leq l \leq r} \left(\frac{\prod_{i=1}^l a_{(i)}}{k} \right)^{2/l},$$

where r is the number of positive widths, listed in decreasing order; set $\psi_a = 0$ if $r = 0$. At $n = 0$ use $H_0 = 1$, also when $s = 1$. The calibration and precision packets belong to the source experiment's actual reports. They do not manufacture two distinguishable future states before the missing information has been acquired.

Let $\mathcal{C}(a, s, L, \delta, b, \varepsilon)$ be exactly this eight-parameter-member family with its stated preparation, report interfaces and task. In the exact-report model, current successful geometric reports may be read as Borel inputs. In the finite-bit model they are accessed through a primitive returning a vector within 2^{-b} in each physical coordinate; the machine may not bypass this primitive. The primitive depends only on the current geometric report and independent randomization, not on σ, η, z ; it is uniform over the family's parameters. Its error guarantee, and finite rational chart data where needed for execution, are part of this second model.

Using the total budget (28), define a single minimax functional

$$(36) \quad \mathfrak{R}_{\mathcal{C}}^X(\mathbf{b}) = \inf_{A \in \mathcal{A}_{\mathcal{C}}^X(\mathbf{b})} \sup_{\lambda \in \{-1, 1\}^2 \times \{0, 1\}} (\mathbb{E}_{\lambda}^A \ell - b_*).$$

The infimum is over parameter-independent machines satisfying the same audited continuation interface and input model. The exact-label model places no finite-bit bound on current Borel comparisons; the finite-bit version below does. In both cases total retained state, and not just a geometric subregister, is constrained by M .

Theorem 6.2 (Joint acquisition–resolution–precision minimax law). *For the family just defined, let $N_{\text{raw}} = n + 3 \geq 3$, $M \geq 12L$, and $b_{\text{num}} = b$. There are positive constants c_d, C_d , depending only on*

d , such that

$$(37) \quad \Phi = H_n V_a + (1 - H_n) \psi_a(\lfloor M/L \rfloor) + \delta^2 + 2^{-2b} + \varepsilon$$

and every nonempty finite-bit or exact-report resource class satisfies

$$(38) \quad \mathfrak{R}_C^{\text{pw}}(\mathbf{b}) \geq c_d \Phi.$$

An exact Borel finite-state encoder with at most M states attains an upper bound $C_d \Phi$. The same upper holds for a finite-bit program on the following explicitly feasible resource region. Suppose the known box endpoints are rational data of description length P_0 , with arithmetic bit length at most B_0 , and the input primitive has the stated error bound. Put

$$m = \lfloor M/(3L) \rfloor - 1, \quad B = B_0 + b + \lceil \log_2(N_{\text{raw}} + M + L + 2) \rceil + 4.$$

A sequential grid scan and schoolbook fixed-precision arithmetic have bounds

$$(39) \quad W \geq C_d B, \quad P \geq P_0 + C_d \log_2(N_{\text{raw}} + M + L + 2) + C_d \log_2(b + 2), \quad \tau \geq N_{\text{raw}} + C_d N_{\text{raw}}(m + 1)B^2.$$

The numerical input precision is b ; arithmetic guard bits and exact rational data are charged in W and P , not supplied as hidden retained state. The calibration input interface must admit the specified common packet. In (39) the time unit charges one raw call and one elementary bit operation; the selected terminal probe is included in N_{raw} . The protocol clock is the fixed call index; arithmetic is padded to the stated uniform bound, and no data-dependent timing is available as a later observation. On this feasible region,

$$(40) \quad c_d \Phi \leq \mathfrak{R}_C^{\text{pw}}(\mathbf{b}) \leq C_d \Phi.$$

The bounds are uniform in the positive widths, rank collapse, $n, s, L, M, \delta, b, \varepsilon$. Taking $L = R_0 D'_0 Q_0$ gives a matched law with growing mandatory simulator/controller/phase continuation memory. All five terms occur in the same family, with one common task and oracle, rather than in unformalized unrelated witnesses.

Proof. For the lower bound, put the uniform prior on the eight values of λ . A supremum risk is at least this Bayes average. Condition on the complete preterminal information and independent algorithm randomness, and apply conditional-mean projection separately to each actually executable terminal probe.

For the geometric block, unsuccessful acquisition leaves X at its prepared law, so its excess over the oracle is at least $H_n V_a$. Conditioned on successful acquisition, its law is still the prepared box law. The tag is independent and must be recoverable exactly. Disjoint tag state supports give integers k_c with sum at most M , even if the machine fuses bit, geometry, clock and controller information. The conditional distribution of X given tag and success is uniform on the same box; all other reports can at most randomize its encoder. The argument of Theorem 6.1 and the anisotropic small-ball bound therefore give at least $c_d(1 - H_n)\psi_a(\lfloor M/L \rfloor)$. Finite current-report precision only restricts this class further and cannot invalidate the lower bound.

Neither preterminal packet nor any other report depends on σ or η . Their posterior means are consequently $1/2$, and their excesses over the same oracle are δ^2 and $h_b^2 = 2^{-2b}/16$. On an erasure the bit posterior under the uniform prior is fair; on a nonerasure it is known. Its conditional variance is therefore $2\varepsilon/4 = \varepsilon/2$. Multiply these blockwise bounds by the common factor $1/4$. The losses add because the block is chosen independently after encoding; no simultaneous counterfactual probe outcomes are assumed. This proves (38).

For the upper, retain the tag, one of the three bit states $0, 1, \perp$, and either the unresolved geometric state or a grid label. Allocate m geometric representatives as displayed. Then $3L(m + 1) \leq M$, and $m \geq M/(6L)$ for $M \geq 12L$. For unresolved geometry predict the exact mean $1/2$; after acquisition round to a valid product grid. The box cover of Lemma 3.2 gives squared error at most $C_d \psi_a(m)$. Holds do not round again. For the two ambiguous probabilities forecast $1/2$. For the bit use its

observed value or $1/2$ on erasure. This rule has the same risk for every λ . Scaling (7) from m to $\lfloor M/L \rfloor$ changes the constant by at most a dimension-independent numerical factor. The exact-label upper follows.

For the finite-bit version, obtain the successful report to coordinate accuracy 2^{-b} and approximate the selected readout to that accuracy. Compute candidate squared distances to absolute accuracy 2^{-2b} using a constant multiple of B guard bits, and minimize these approximations. If r is the closest distance to the quantized input, the selected distance is at most $\sqrt{r^2 + C_d 2^{-2b}} \leq r + C_d 2^{-b}$. Including the input error gives its covering radius plus $C_d 2^{-b}$. Its squared error is at most twice the covering error squared plus $C_d 2^{-2b}$. The unresolved mean and the three bit forecasts are dyadic and exact. The added acquired-report error is absorbed by the already present 2^{-2b} term, without an inverse width.

For completeness, choose grid cell counts by the finite dyadic-scale construction in Appendix A, rather than by comparing arbitrary real algebraic minima. This is a constant-factor cover with at most m points. Iterate through its mixed-radix indices to generate each center on demand. Store a current index, the best index, the tag and bit state, and $O(d)$ current B -bit coordinates. Only the chosen index, tag and bit state survive the call. Rounded schoolbook multiplication, division and distance comparison take at most $C_d B^2$ bit operations per candidate, with guard bits ensuring the stated absolute error. Even scanning at every call costs no more than the time in (39); in fact only the successful report needs a scan. The readonly program holds the rational data and binary descriptions of its budgets, not an observation-dependent table. This proves the resource bounds and the implementable upper. Zero widths are processed as exact point coordinates, never as divisors, so the constants survive their disappearance. $\square$

Proposition 6.3 (Exact erasure deficiency and the two elementary floors). *In the declared parameter-independent simulator class, the deficiency of the bit-erasure experiment with erasure probability θ relative to the exact-bit experiment is exactly $\theta/2$. With a uniform bit preparation its best squared-prediction risk is $\theta/4$. For two identical preterminal calibration reports and terminal probabilities $1/2 \pm h$, the minimum excess over the sign-knowing oracle under the equal prior is h^2 . These statements use the same squared loss as Theorem 6.2.*

Proof. Write P_0, P_1 for the source report laws. Their total variation distance is $1 - \theta$. Any parameter-independent simulator K contracts total variation. If both simulated target laws are within e of the exact target laws δ_0, δ_1 , the triangle inequality gives

$$1 = \|\delta_0 - \delta_1\|_{\text{TV}} \leq 2e + \|KP_0 - KP_1\|_{\text{TV}} \leq 2e + 1 - \theta.$$

Thus $e \geq \theta/2$. Filling an erasure by an independent fair bit attains this bound for each parameter. Including the actual terminal bit in the comparison law changes neither conclusion: the marginal lower persists, and under either fixed parameter the fair-imputation error event has probability exactly $\theta/2$. The posterior variance under the uniform preparation is $1/4$ on an erasure and zero otherwise, proving the risk assertion. For the calibration pair the conditional mean is $1/2$ and conditional-mean projection gives the excess $\mathbb{E}(p - 1/2)^2 = h^2$, attained by that forecast. $\square$

Theorem 6.2 is a full-ledger risk assertion on its stated feasible region, not a universal lower bound for the shortest program or fastest arithmetic in every kernel class. Its total-memory converse does not require constant overhead, but does require the audited continuation interface. The b -indexed information cell is compulsory in this experiment: optional inaccurate arithmetic would not imply its lower floor. These distinctions are part of the theorem's quantifiers, not qualifications inserted after changing the task.

7. RAW-KERNEL REALIZATION THEOREMS

7.1. A controlled binary hidden-state filter. Let $X_t \in \{0, 1\}$ have transition probability $q_0 \in (0, 1/2)$ of flipping at each raw call. The preparation is $P(X_0 = 1) = 1/2$. A command selects

a from a fixed finite subset of $[a_-, a_+] \subset (0, 1)$. Conditional on the new hidden state, the report $y \in [-1, 1]$ has density

$$(41) \quad g(y \mid X_{t+1} = x, a) = \frac{1}{2}(1 + a(2x - 1)y).$$

Each call costs one acquisition and one unit of physical time. The exploration may be any specified causal command rule, fixed independently of the encoder. A terminal probe reads X_N and ends the experiment. Further filtering reports and this terminal probe are the designated future tests.

Theorem 7.1 (Filtering realization). *For this raw-kernel class, the predictive quotient is the scalar conditional probability $p_t = P(X_t = 1 \mid H_t)$, with the terminal/continuation interface. It has a common compact invariant interval after initialization. For all $N \geq 1$ and $m \geq 1$,*

$$(42) \quad R_{\text{cp}}(N, m) - B_N \asymp R_{\text{on}}(N, m) - B_N \asymp m^{-2}, \quad B_N = \mathbb{E}[p_N(1 - p_N)].$$

The constants depend only on q_0, a_-, a_+ , not on N or on the specified command rule. The zero-error upper is an exact Borel finite-label implementation. With rational or certified finite-precision calibration, the separate program in Appendix A has the following nonzero-error guarantee. A uniform update/numerical error u in log-odds distance contributes at most $u/(2q_0)$ to the propagated state error. The constants in this statement are not asserted uniform as $q_0 \downarrow 0$.

Proof. Direct normalization of (41) gives, with $r = q_0 + (1 - 2q_0)p$,

$$(43) \quad f_a(y \mid p) = \frac{1}{2}[1 + ay(2r - 1)], \quad F_{a,y}(p) = \frac{r(1 + ay)}{1 + ay(2r - 1)}.$$

All denominators are at least $1 - a_+$. Conditional laws of all future report words and the terminal probe are functions of p by induction. Conversely the terminal probe has mean p , so different p are predictively different. This proves quotient identification and minimality without assuming a covariance or information matrix.

Let

$$c_- = \frac{q_0(1 - a_+)}{1 + a_+(1 - 2q_0)}, \quad c_+ = 1 - c_-.$$

The interval $D = [c_-, c_+]$ contains $1/2$ and is invariant, because the prediction r always lies in $[q_0, 1 - q_0]$ and the update is increasing in r and ay . Use the metric $d(p, p') = |\text{logit}(p) - \text{logit}(p')|$. On D it is equivalent to Euclidean distance, with constants depending only on the displayed parameters. The map $p \mapsto p$ is the task map, and its inverse-coordinate Lipschitz constants follow from the bounded derivative $1/[p(1 - p)]$. This provides a one-dimensional global chart and an effective valid domain.

For update stability, put $v = p(1 - p)$ and $\rho = 1 - 2q_0$. The derivative of the predicted log-odds with respect to previous log-odds equals

$$\frac{\rho v}{q_0(1 - q_0) + \rho^2 v} \leq \rho.$$

Indeed $v \leq 1/4$ and $1 - \rho^2 = 4q_0(1 - q_0)$, so the denominator is at least v . The observation then adds $\log[(1 + ay)/(1 - ay)]$, independently of the previous state. Thus every same-report update is ρ -Lipschitz in d . Proposition 3.4(a) gives $\Gamma_N \leq (2q_0)^{-2}$, including initialization rounding.

It remains to verify actual acquired mass. For each preceding history and command,

$$(44) \quad \frac{\partial F_{a,y}(p)}{\partial y} = \frac{2ar(1 - r)}{[1 + ay(2r - 1)]^2} \geq c_y := \frac{2a_-q_0(1 - q_0)}{(1 + a_+)^2} > 0.$$

The conditional density of y is at most $(1 + a_+)/2$. Change of variables therefore gives a conditional density of p_{t+1} at most $(1 + a_+)/(2c_y)$ on its conditional interval and zero elsewhere. Mixtures over the actual histories and commands preserve this upper bound. Its integral is one. Consequently (6) holds on the fixed interval D at every $N \geq 1$, with C_g this bound times $|D|$. No uniform posterior law has been inserted. Equations (5) and (6) give $\Psi_N(m) = |D|^2/m^2$.

For report continuity, direct integration of the density difference gives

$$\|f_a(\cdot | p) - f_a(\cdot | p')\|_{\text{TV}} = \frac{a\rho}{2}|p - p'| \leq \frac{a+\rho}{8}d(p, p').$$

The terminal Bernoulli kernel has total variation $|p - p'| \leq d(p, p')/4$. Hence all loss-relevant reports satisfy (C). An equally spaced grid in p , the update (43), and exact Borel threshold comparisons define the mathematical label encoder. This is not yet a finite-bit implementation. For rational calibrated constants, Appendix A gives the separate finite-bit routine, including current-report quantization, interval arithmetic and absolute-error conversion. Approximate input and arithmetic with certified absolute errors are absorbed into u_t ; denominator and log-odds conversion bounds above translate them to the common metric. For rational calibration, fixed-precision interval arithmetic produces any prescribed positive update accuracy; an exact tie need not be resolved, since approximate nearest-grid selection suffices. The persistent datum is the grid index; a bounded-precision register for the current report and arithmetic is temporary, not a retained real state. Summing the geometric error series gives the last error assertion. Apply Theorem 2.1 to conclude. $\square$

This is a repeatedly updated nonlinear filter, not a once-prepared finite measurement ensemble. The acquired law is generated by the actual observation density at every positive time. Formula (42) concerns prediction along the selected exploration, not the unrestricted optimal information-gathering controller.

7.2. Singular geometric acquisition with an absorbing report. We next construct a class with no strict contraction during waiting and with genuinely intersecting, different-dimensional acquired supports. Fix finitely many compact boxes Q_j and maps

$$z_j : Q_j \longrightarrow [1/4, 3/4]^d$$

that are bi-Lipschitz with uniform constants in physical Euclidean coordinates. They may be nonlinear and their images may intersect. A dimension-zero box is a point. Prepare a latent mark J_0 with probabilities π_j and, conditionally on $J_0 = j$, a point V uniformly on Q_j . The hidden target is $Z = z_{J_0}(V)$. After preparation the raw hidden state is (Z, A_t) ; the latent mark is retained only on the preparation probability space for analyzing its pushforward law. The mark and any zero-width source coordinates are not supplied to the observer.

The apparatus has an unresolved/absorbed flag $A_t \in \{0, 1\}$, initially zero. At each raw attempt, when unresolved, an independent coin with success probability $s \in [0, 1]$ is tossed. Failure reports $\perp$ and changes nothing. Success reports Z and sets the flag to one. Thereafter every raw attempt reports $\perp$ and preserves Z and the flag. All attempts, including these holds, cost one acquisition and one unit of physical time. The action “attempt” is available in both modes, so no unrecorded action-legality bit is assumed. The terminal menu consists of a flag probe returning A_N , and d binary probes with success probabilities Z_i , each executed only once as a terminal experiment. Give these $d + 1$ probes equal weights.

After failure while unresolved, the hidden target distribution is unchanged: failure probability is independent of Z . After a successful report the future target probabilities are exactly the reported Z , whether or not the latent mark is identifiable. Thus the predictive quotient is the unresolved point o , together with the *glued* set of absorbed predictions $z \in \bigcup_j z_j(Q_j)$. At an intersection the mark is quotiented out, not secretly retained. The flag probe distinguishes o from an absorbed state even when their coordinate means coincide.

Set

$$(45) \quad b_* = \frac{1}{d+1} \sum_{i=1}^d \mathbb{E}[Z_i(1 - Z_i)], \quad V_* = \frac{1}{d+1} \sum_{i=1}^d \text{Var}(Z_i), \quad h_N = (1 - s)^N.$$

For $N = 0$ take $h_0 = 1$, including $s = 1$. Let $\Psi^{\text{acq}}(m)$ be (7) for the absorbed component charts with weights π_j . Widths there are the actual physical chart widths with the fixed task-weight scaling absorbed into the chart constants.

Theorem 7.2 (Stratified acquisition law through rank collapse). *For every $s \in [0, 1]$, $N \geq 0$, and $m \geq 2(J + 1)$,*

$$(46) \quad R_{\text{cp}}(N, m) - b_* \asymp R_{\text{on}}(N, m) - b_* \asymp h_N V_* + (1 - h_N) \Psi^{\text{acq}}(m).$$

The constants depend only on the finite chart and task constants, not on s, N , positive widths, or the distance between intersecting strata. When positive widths vanish, zero coordinates are removed from the predictive quotient and the same assertion holds, including point components. If all component score images (including the unresolved point) are separated, Theorem 2.1 gives the all-budget version using the full mixture allocation (7), for every $m \geq 1$.

The online construction has $\Gamma_N = 1$ in the full-mixture transport certificate. It retains no success count or latent component mark. Finite-precision successful reports with metric error at most η give a root-mean-square prediction allowance at most $C\sqrt{1 - h_N}\eta$, when all hold operations are exact.

Proof. The preparation is the pushforward of the specified box mixture, and the transition is a normalized nonnegative Borel kernel on the compact target space with its two flags. For the success report component one may take the prepared target law as dominating measure; its support is the compact union of the chart images. The success instrument at the unresolved belief has density s , and its determining-test numerator has the continuous version $sf(z)$. At absorbed beliefs both densities are zero. The $\perp$ component is atomic. Thus the explicit overwrite below provides exactly the fibre-compatible zero-density extension required in Proposition 1.2. The admissible belief domain is the unresolved prepared law together with the point beliefs at absorbed targets. It is a disjoint compact union. All determining future-test evaluations are continuous on it; the explicit overwrite/hold versions below also verify the zero-denominator extension clause of Proposition 1.2. The probability of N consecutive failures is exactly h_N . Conditional on success by N , the target law is still the prepared mixture, because all acquisition coins are independent of it. The acquired predictive law is therefore

$$(47) \quad h_N \delta_o + (1 - h_N) \sum_j \pi_j (z_j)_\# \text{Unif}(Q_j),$$

with the absorbed flag coordinate included in the score. In particular the formal union of all prepared values does not have weight one before it is acquired.

The conditional terminal variance of the flag is zero. On an absorbed history the coordinate variance is $Z_i(1 - Z_i)$. On an unresolved history it is $\mathbb{E}[Z_i(1 - Z_i)] + \text{Var}(Z_i)$, by total variance. Averaging proves

$$(48) \quad B_N = b_* + h_N V_*.$$

Conditioned on the raw component mark, the acquired charts have exactly uniform physical box densities. Their bi-Lipschitz constants give (G). Their union can have intersections, different local dimensions, and atomic mass. All these properties follow from the raw preparation and report map.

Equip the quotient with Euclidean distance on its weighted score vector. Report continuity holds as follows. Between two absorbed states the next-attempt kernels are identical, namely $\delta_\perp$. Between unresolved and absorbed modes, the variation distance is at most one, while the flag-score distance is $1/\sqrt{d+1}$; a finite global Lipschitz constant follows. Terminal coordinate kernels are Bernoulli and their variation distance is $|Z_i - Z'_i|$; the flag kernel is bounded by the same mode-separation argument. There is no assertion that distinct Dirac reveal channels are TV-continuous: the reveal action is used only from the single unresolved quotient point. On a successful report define the predictive update to overwrite with that reported absorbed state from every previous representative, including the conditional version at a null report in the absorbed mode. On $\perp$, it is the identity.

The machine initially encodes o . At a success it chooses a nearest representative in the union codebook, using the reported physical coordinates; on every failure or later hold it leaves the label unchanged. It never reads J_0 . Conditional on any raw mark j , the success-rounding error is bounded by that chart's cover. Therefore the coefficients are exactly those of Proposition 3.4(c), with the unresolved point as component zero. This proves $\Gamma_N = 1$, irrespective of how many holds occur. It also proves the finite-precision allowance: a single error of size $C\eta$ is introduced on an event of probability $1 - h_N$, and none on the complementary event.

For completeness one can see the separated oracle and acquisition terms directly. Lemma 3.1, conditioned on the success event and its latent mark, and the proof of Corollary 3.5 give an excess of at least $c(1 - h_N)\Psi^{\text{acq}}(m)$, even if the decoder also uses centers for the unresolved point. For the upper use one exact unresolved label and $m - 1$ absorbed grid labels. Allocation scaling gives $\Psi^{\text{acq}}(m - 1) \leq C_J \Psi^{\text{acq}}(m)$ for $m \geq 2(J + 1)$: allocating $\lfloor (m - 1)/J \rfloor$ to every chart and using the estimate in Corollary 3.5 proves it. The finite machine just constructed then incurs at most $C(1 - h_N)\Psi^{\text{acq}}(m)$. Add (48). At $s = 0$ or $N = 0$ the success term is zero and a single known forecast attains the acquisition term. With zero surviving widths all component charts are points and sufficiently many labels give zero acquired distortion, as the formula states.

For effective implementation, take algebraic calibrated widths and weights, and chart maps with certified finite-precision evaluation. The finite allocation minimum is an algebraic comparison problem. A nearest representative can be selected to within absolute distance η by comparing approximated distances; this adds at most 2η to the best covering radius. No exact comparison of arbitrary input reals is required. A finite readonly table of approximated representatives or a grid-generation program, with separately charged description and temporary workspace, suffices. Thresholds are compared in physical coordinates; no uniform error estimate divides by a collapsing width. If the unresolved target mean also requires numerical integration, its certified score error η_0 is charged separately; the total readout allowance is then at most $\sqrt{h_N}\eta_0 + C\sqrt{1 - h_N}\eta$. A table or algorithm for this mean is part of the supplied effective calibration, not a free integration oracle. Appendix A distinguishes these costs. This proves the claimed implementation statements. $\square$

A concrete singular family. In two coordinate dimensions take equal mixture weights on

$$z_1(u) = (1/2 + u, 1/2), \quad |u| \leq a/2, \quad z_2(v, w) = (1/2 + v, 1/2 + w), \quad |v| \leq b/2, |w| \leq c/2,$$

where $0 \leq a, b, c \leq 1/4$. The actual acquired law has a one-dimensional singular component on a line and a two-dimensional component on a rectangle. The former lies inside the latter when their horizontal ranges overlap. If $c = 0$, the rectangle becomes a line; if also $b = 0$, it becomes an atom. Before acquisition the support consists only of the unresolved predictive point. The physical scales are

$$\psi_1(k) = a^2/k^2, \quad \psi_2(k) = \max\{\max(b, c)^2/k^2, bc/k\} \quad (bc > 0),$$

with zero widths removed at the boundary. There is no single smooth acquired manifold at the crossing or a single ambient-dimension exponent. Theorem 7.2 applies with constants uniform through all these changes. Smooth nonalgebraic bi-Lipschitz distortions of the two charts yield the same conclusion, with their uniform metric constants. Thus the extension is not a statement about a preassigned Hilbert-space rank cut.

7.3. Recurrent expansion with intermittent refresh. We give a third realization to test the structural kernel theorem, rather than only its contractive and absorbing special mechanisms. Fix $l = 1/4$, $0 \leq a \leq 1/2$, and $u = l + a$. Prepare X_0 uniformly on $[l, u]$; for $a = 0$ this is a point. One initial paid call reveals X_0 . At each subsequent paid tick, independently of the past, a hold occurs with probability $1 - \theta$, $0 \leq \theta \leq 1$. On a hold, X is unchanged and the report is the symbol H . Otherwise an active type is chosen independently: with probability $p = 1/10$ it is the fold E , and

with probability $1 - p$ it is the refresh C . The raw transition/report kernels are

$$(49) \quad \begin{aligned} E: \quad X' &= l + 2 \min\{X - l, u - X\}, \quad Y = E, \\ C: \quad X' &= cX + (1 - c)(l + aU), \quad Y = (C, U), \quad c = 1/4, \quad U \sim \text{Unif}[0, 1]. \end{aligned}$$

These are normalized nonnegative Borel kernels; the fold is deterministic. There is no further exact reveal of X' . A terminal probe is Bernoulli with mean the current X . All ticks, including the initial reveal and holds, cost one raw call. The exploration has no choice of informative actions and is fixed. Its protocol clock is exogenous.

Theorem 7.3 (Acquired geometry under recurrent expanding updates). *For every $N \geq 1$, $m \geq 1$, $a \in [0, 1/2]$ and $\theta \in [0, 1]$, the raw experiment (49) has predictive quotient $[l, u]$ with its protocol interface, and*

$$(50) \quad R_{\text{cp}}(N, m) - B_N \asymp R_{\text{on}}(N, m) - B_N \asymp a^2 m^{-2}, \quad B_N = \mathbb{E}[X_N(1 - X_N)].$$

Here X_N denotes the state after the N paid calls, including the initial reveal. The constants are absolute and uniform in all displayed parameters. At $a = 0$ both exact excesses are zero. The online bound has the kernel-derived transport constant

$$(51) \quad \Gamma \leq \frac{3040}{667},$$

including initial rounding, independently of the hold probability.

If current inputs, physical endpoints and arithmetic have absolute errors controlled by η per initial or active update, and hold operations are exact, the finite-bit implementation has

$$R_{\text{alg}}(N, m) - B_N \leq C(a/m + \eta)^2.$$

The upper is independent of N and θ . The endpoint/error guarantee includes a projection back to the calibrated valid interval with its stated absolute accuracy. Its program and workspace costs are those of Appendix A; no stored exact real is used.

Proof. After the initial report the exact state is known. The reported type and, at refresh, the reported U , determine each subsequent exact update. All future laws are therefore functions of the current x and the protocol interface. The terminal Bernoulli mean is x , so distinct states are predictively distinct. Continuation laws and these explicit updates are continuous on the interval within each report component. The report law before the terminal probe is independent of x , hence has zero report-kernel continuity constant. The terminal report has total variation $|x - x'|$. This establishes the quotient and report certificates directly from the raw kernel, without assuming a covariance geometry.

For $a > 0$, let f be the actual current density. Initially $f = 1/a$. A hold preserves it. The tent fold has two affine inverse branches, each with Jacobian factor $1/2$, so its pushforward density is the average of the two branch densities and has sup norm at most $\|f\|_\infty$. For refresh, condition on the preceding state. The new state is uniform on an interval of length $(1 - c)a$; its conditional density is at most $1/((1 - c)a) = 4/(3a)$. Mixing over the actual preceding law preserves this bound. Induction, including mixing over the reported type and holds, gives

$$0 \leq f_N(x) \leq 4/(3a), \quad \int_l^u f_N(x) dx = 1$$

for every $N \geq 1$. Thus the actual-mass certificate has $C_g = 4/3$ on one chart of physical width a . It is a bound on the law generated by the experiment, not an imposed invariant uniform law. The task map is $x \mapsto x$ and $\psi(m) = a^2/m^2$.

Use a midpoint grid with m representatives and physical Euclidean distance. Execute holds without arithmetic or rounding. Execute an active update from the previous representative and then round. The fold has Lipschitz constant two; refresh has constant $c = 1/4$. Their rounding error is at most $a/(2m)$. Therefore the error recurrence has active gain $G = 2$ with probability $1/10$ and

$G = 1/4$ with probability $9/10$, with forcing one after scaling the grid radius. The coefficients are independent marks, independent of the grid. In the one-state environmental kernel,

$$\mathbb{E}G = 17/40, \quad \mathbb{E}G^2 = 73/160 < 1.$$

Theorem 4.1 and (26) yield

$$H = \frac{1 + 17/40}{(1 - 17/40)(1 - 73/160)} = \frac{3040}{667}.$$

The normalized initial rounding seed is at most one; the scalar fixed-point first and second moments are both at least one, so it is an admitted seed. Proposition 4.3 proves that insertion of holds leaves this bound unchanged for $\theta > 0$. At $\theta = 0$ only the initial rounding persists, so the same upper holds directly. Apply Lemma A.1 to the one paid initial reveal and the $N - 1$ subsequent calls. Theorem 2.1 then supplies (50), and its lower uses the actual density just proved. No realization calculation has entered the proof of the kernel theorem.

On an active or initial update, interval projection and approximate arithmetic add at most $C\eta$ to the same error recurrence. On holds the representative is copied exactly, giving zero new error. The same scalar forcing bound, applied to radius $a/(2m) + C\eta$, proves the finite-bit assertion. The physical formulas in (49) use a only multiplicatively; their absolute-error bounds never divide by it. At $a = 0$ there is a single exact state, and the exact program can bypass all rounding. This also proves rank-uniformity. $\square$

For every $\theta > 0$ the active folds occur repeatedly, and blocks of consecutive folds have positive probability. Thus this example is not uniformly contractive along active paths, not an absorbing reveal after the initial preparation, and not a finite measurement ensemble. Its stability is a forced second-moment conclusion derived from the raw update probabilities. Uniformity in arbitrarily long waiting periods does not remove those periods from the acquisition or physical-time budget.

8. CONSEQUENCES AND COMPARISON WITH EXISTING THEORY

The acquired law, its geometric resolution and the propagation of implementation error are three different objects. They are coupled here by a common experiment and task, not by an ambient parameter dimension. Theorem 2.1 makes this coupling quantitative. Theorem 4.1 supplies its causal estimate from one-step marked-kernel data; the normalized matrix in (24) records, rather than suppresses, the remaining comparison with acquired mass. Theorem 6.2 then gives a single fully specified experiment class in which acquisition, memory and compulsory information losses occur together.

The three realizations test distinct parts of this mechanism. The filter derives a posterior density from the actual report Jacobian. The absorbing sensor derives a mixed-dimensional acquired measure and a disjoint-indicator transport identity. The recurrent experiment derives a density bound under repeated folding and refresh, and has active expansions even though its forced second moment is bounded. Each follows the general theorem; none is a premise of it.

8.1. Theorem-level comparison.

Predictive states and sufficient statistics. Shalizi and Crutchfield's minimality and uniqueness theorems for causal states [9, Theorems 2–3] identify minimal prescient representations of a stochastic process. Predictive state representations [5] use future action–observation tests. Our quotient proposition does not claim priority for either construction. It explicitly includes controlled continuation legality, paid resources and conditional-version domains. Its extra measurable work is the density-instrument descent at every declared continuous report. The categorical treatment of kernels and sufficiency in [15] supplies another general language; an abstract sufficient statistic alone does not provide the acquired-mass or finite-machine estimates used here.

Bisimulation and belief-state control. Behavioral metrics for Markov decision processes [10] compare states through rewards and transition laws and control value differences. Our metric is fixed by executable prediction tasks and valid update geometry; it is not asserted to be a universal bisimulation metric. Saldi, Yüksel and Linder [11, Assumption 1 and Theorem 3] obtain near-optimal finite belief-model policies for discounted partially observed control under their continuity, compactness and moment hypotheses. That theorem optimizes a controller, whereas our matching lower is for a fixed exploration and the law it actually acquires. Our finite-horizon two-sided resolution estimate does not imply their unrestricted control conclusion, nor does asymptotic policy near-optimality supply our actual-mass lower bound.

Quantization and nonlinear filtering. Conditional-mean projection and optimal probability quantization are classical [4]. Nonlinear filter approximation by quantization is developed in [6, 7, 3]. The additional obligation here is a matching lower for the very same acquired task, together with the forced-moment condition needed by a persistent finite machine. Zhu’s local estimates for Ahlfors–David measures [8, Theorem 1.1] concern a substantially broader singular-measure setting than a finite union of box charts. Our finite-budget anisotropic bounds expose constants through width collapse and intersecting finite mixtures; they neither classify arbitrary singular measures nor replace established quantization-dimension theory.

Positive moment operators. Average contraction of iterated random functions [2] and moment stability of Markov jump systems [12] are established subjects. We claim no novelty for a Neumann resolvent or the criterion $\rho(K_2) < 1$. The role of Theorem 4.1 is more specific: reverse-edge operators propagate the conditional first and mixed second moments of the *rounding forcing*, and the resulting matrix is normalized by the actual acquisition weights. Exact suspension cancels in the forced profiles, not in the physical budget. This distinction is why a recurrent expanding kernel can still yield a matching finite-memory risk curve.

Comparison of experiments. Blackwell comparison [1], Le Cam’s deficiency theory [13], and Torgersen’s systematic account [14] provide the classical common-decision-loss principle. The triangle inequality and total-variation contraction in Theorem 5.1 are instances of that principle. The added structure is the finite-resource causal implementation, with explicit lifted controller classes, stopping indices, clocks and numerical interfaces. A forward simulator yields an upper risk transfer, not a lower bound relative to a different oracle. The mandatory-state and robust converses are proved directly in Theorems 6.1 and 6.2, rather than attributed to a forward deficiency bound.

8.2. Necessary distinctions illustrated by exact examples. First, the valid envelope $[0, 1]$ equipped with a point mass and the same envelope equipped with a uniform law have respectively zero one-point distortion and strictly positive finite-label distortion. Hence a reachability envelope is not an acquired law. More generally a component of probability w contributes with weight w , even if its geometric diameter is large.

Second, an identity update has no need for repeated quantization. If the initial state has already been replaced by a representative, the same representative can be retained indefinitely. The bound Nr obtained by blindly injecting a covering radius at every hold is not an inherent resource requirement. The acquisition-weighted certificate corrects precisely this failure of accounting.

Third, a statistical garbling can increase memory needed to match its *own* oracle. Observing a fair hidden bit exactly produces only posterior probabilities zero and one. Adding independent Gaussian noise to this observation produces a continuously varying posterior probability. Two labels reproduce the first oracle exactly, but no finite number reproduces the latter continuously distributed oracle exactly under square score. This is compatible with data processing because the baselines differ.

Fourth, let $P_N = \delta_0$ and $Q_N = (1 - e^{-Nc})\delta_0 + e^{-Nc}\delta_1$, $c > 0$. Their total variation distance tends to zero, but the exponential rates at the point one are infinite and c , respectively. The common-risk defect of this paper is therefore not a certificate of large-deviation equivalence.

Finally, a hidden mode or an internal simulator register is not free because its name is omitted from a state vector. Independent pairs of retained states have the product cardinality in Theorem 5.1. Conversely that product is a constructive upper bound, not a theorem that every simulator requires it. Theorem 6.1 establishes necessity only through separately executable, zero-error continuation challenges. An exact label machine and a finite-bit numerical implementation are also different objects: the latter retains the nonzero calibration, workspace, and readout terms of (32). Exact-arithmetic zero distortion is not a promise of a finite-precision physical output with zero error.

8.3. Uniformity and the remaining mathematical questions. The conclusions apply to the experiment classes actually specified. For the structural route, a marked factor, invariant environment law, dominating preparation, bounded Lyapunov weight and acquired-weight normalization must be verified. In the one-chart recurrent class the normalization is automatic, and active expansion is permitted. In many-chart classes it can fail when a terminal component becomes rare while earlier innovations still transport into it. A sharper finite-horizon localization is then a mathematical problem, not a free consequence of the resolvent bound.

The total-resource converse is nonconstant in the mandatory continuation memory. It does not establish a shortest-program, minimum-workspace or optimal-runtime converse below the constructive feasible region, and a small allowed tag-error probability would require a quantitative relaxation of the continuation-separation argument. The robust precision floor comes from compulsory indistinguishable input cells; optional numerical inaccuracy has no such converse. These are different proof responsibilities even though all coordinates are retained in the upper resource map.

The framework also leaves concrete extensions to be proved: learning unknown kernels with optimally charged pilots; unrestricted adaptive exploration with a common lower-bound experiment; infinitely accumulating strata beyond the finite-chart mass bounds; and adaptive or state-dependent suspensions not covered by Proposition 4.3. Bounded stopping comparisons are proved, but an infinite-horizon transcript comparison, a large-deviation equivalence or closure for unbounded generators would require stronger and different estimates. None is inferred from finite-time total variation or from a realization-specific identity. The central program remains the same experiment-to-geometry-to-resource transfer, rather than a relabeling of any single realization.

APPENDIX A. INITIALIZATION, EXACT ENCODERS AND EXECUTABLE APPROXIMATIONS

This appendix specifies which operations the finite programs perform. A finite set of mathematical labels does not, by itself, certify finite-bit computation. Conversely, a physical program need not solve an exact-real nearest-neighbor tie in order to attain a covering bound.

Lemma A.1 (Paid initialization and continuation). *Suppose a raw initialization protocol uses n_0 calls, has a legal online encoder within the declared state capacity, and produces an observed record H_0 , an exact predictive state in the valid continuation domain and an initial representative. Suppose the geometric, local update and transport hypotheses are verified under the joint actual initialization and continuation law, including the initial error in (I). Then the proof of Theorem 2.1 applies to T subsequent calls, with $n_0 + T$ preforecast calls and total raw budget $n_0 + T + 1$ when one terminal physical probe is executed and the initializer’s workspace, program, clock and physical-time costs included. No conditioning on a successful initialization is permitted unless its probability and failure branches are retained.*

Proof. The projection identity conditions on the complete record (H_0, H_T) and does not require H_0 to be deterministic. The geometric lower integrates its actual terminal law. The online proof begins with the stated initial representative error and unrolls exactly the same recurrence for the T subsequent updates. Concatenating the two finite machines is legal by the initializer hypothesis and the common state interface. Its raw calls and physical time add. This proves the assertion without postselection or an unpaid initial observation. $\square$

A.1. A width-uniform computable grid. For a box with rational physical widths and a label allocation $k \geq 1$, remove zero coordinates. If none remain, use its one point. Otherwise let $a_{\max}$ be its largest width and try scales $R_s = a_{\max} 2^{-s}$, beginning with $s = 0$. Set $n_i(s) = \max(1, \lceil a_i/R_s \rceil)$ and stop before the first scale with $\prod_i n_i(s) > k$. Multiplication may be capped at $k+1$. Output the midpoint grid of the last feasible scale. These are comparisons of specified rational data. There are at most $2 + \lfloor \log_2 k \rfloor$ trials (including the rejected scale) because the first coordinate has $n_1(s) = 2^s$.

If $h = \sqrt{\psi(k)}$, the proof of Lemma 3.2 shows feasibility at scale $2h$. Some tried scale between $2h$ and $4h$ is therefore feasible, unless the initial one already has smaller width. The chosen grid has radius at most $2\sqrt{d_*}h$ and at most k points. No inverse disappearing physical width enters this error constant. Exact comparisons may require more input bits when a nonzero width is extremely small; those bits belong to the rational description length, not to an allegedly uniform free oracle. A finite mixture may use exact algebraic allocation search, or any certified constant-factor allocation; these are separate clauses in the main theorem.

An approximate nearest point is selected as follows. Read the current physical report to norm accuracy η . Generate candidate centers on demand to accuracy $O(\eta^2)$ and compute their squared distances to absolute accuracy $O(\eta^2)$. Retain a least computed distance, breaking ties by index. If r is the best true distance from the quantized report, the chosen distance is at most $\sqrt{r^2 + C\eta^2} \leq r + \sqrt{C}\eta$. Adding the input error gives covering radius plus $C'\eta$. This algorithm compares rational approximations and never has to certify inequality of arbitrary real oracles.

A.2. Exact Borel transitions. The following is pseudocode for mathematical label machines. The current real report is a transient Borel input, not a real number retained between calls.

```
FILTER(label, command a, current report y):
  p := valid_grid_point(label)
  r := q0 + (1 - 2*q0)*p
  p_new := r*(1 + a*y)/(1 + a*y*(2*r - 1))
  return first_nearest_grid_label(p_new)
```

```
ABSORBING_SENSOR(label, current report):
  if report is the hold/failure symbol: return label
  return first_nearest_valid_union_label(report)
```

```
FOLD_REFRESH(label, current report):
  if report is H: return label
  x := valid_grid_point(label)
  if report is E: x_new := l + 2*min(x-l, u-x)
  if report is (C,U): x_new := x/4 + 3*(l+a*U)/4
  return first_nearest_grid_label(x_new)
```

The sensor starts with its unresolved label; its latent chart mark is never read. The recurrent encoder begins by encoding the paid initial reveal into the same grid. Before that reveal its protocol phase is fixed and requires only one known state. Lemma A.1 then applies with $n_0 = 1$; N in Theorem 7.3 is the total $1 + T$, not an uncharged continuation horizon.

A.3. Finite-precision versions and local errors. For a finite-bit machine replace the exact report by its certified numerical input, evaluate the update with interval or rounded rational arithmetic, project to the declared valid domain, and use the approximate nearest-point rule. A copied hold is an exact integer label copy. No arithmetic is done on a hold, so its innovation and local numerical error are zero. The following local error statements quantify the change from the exact Borel machine.

For the filter, work in a fixed safe calibrated parameter rectangle

$$q_0 \in [\underline{q}, \bar{q}] \subset (0, 1/2), \quad a \in [a_-, a_+] \subset (0, 1).$$

On it the denominator in (43) is at least $1 - a_+$, and the invariant probabilities stay a positive distance from zero and one. The update is a rational function with all first derivatives in (p, q_0, a, y) bounded on this compact domain. The derivative of logit is bounded on its image. By the mean-value theorem, parameter error at most δ , report error at most η , and arithmetic absolute error at most η give

$$(52) \quad u_t \leq C_{\underline{q}, \bar{q}, a_-, a_+}(\delta + \eta), \quad v_N^{\text{num}} \leq C\eta.$$

Round/clip approximate calibration into the safe rectangle. A common valid representative interval may be enlarged using the rectangle bounds; all derivative and density constants remain controlled by these displayed parameters. Consequently the propagated state error is at most $C(\delta + \eta)/(2\underline{q})$. Choosing $\delta + \eta = O(m^{-1})$ recovers the m^{-2} risk order with finite bits. Exact calibration and exact-real comparisons are not described as a finite-precision program.

For the singular sensor use physical target coordinates. A successful numerical report within η , an approximate nearest valid center and a readout within η contribute at most $C\eta$ once. Let η_0 bound a precomputed unresolved mean. The root-mean-square readout allowance is

$$(53) \quad \sqrt{h_N \eta_0} + C\sqrt{1 - h_N \eta}.$$

A finite table or a supplied certified chart-evaluation and mean-computation routine must be included in P . Computability of a chart map without a complexity bound on its mean integral does not give a universal complexity bound. For affine rational boxes the mean is explicit. No approximation in (53) divides by a collapsing width or identifies the latent chart mark.

For the recurrent sensor the maps min and affine interpolation are Lipschitz in physical coordinates. Endpoint, report and arithmetic errors bounded by η therefore give $u_t \leq C\eta$ on the initial or active steps and $u_t = 0$ on holds. The same forcing profile used for the grid error propagates this numerical error, yielding the last assertion of Theorem 7.3. Calibration is either exact rational data or an input with the stated uniform absolute-error guarantee; no unknown endpoint can be read without it.

A.4. Separate storage and time accounts. With a procedurally generated grid and rational data, a one-dimensional filter or recurrent sensor needs a persistent index of $\lceil \log_2 m \rceil$ bits and $O(B + \log m)$ transient bits for current reports and arithmetic, where B includes calibration input length, numerical accuracy and guard bits. A scan costs at most $O(mB^2)$ bit operations per active step; a uniform one-dimensional grid also permits a direct rounded-index calculation. The description stores the rational constants and budget integers, not a table of m arbitrary exact reals. For a generic supplied union chart table of m centers in d coordinates at B bits, readonly data instead cost $O(mdB)$ and the distance scan has its own time cost. These are alternative implementations, not interchangeable accounts.

The audited-tag minimax program additionally retains the tag and one three-valued bit state. Its geometric unresolved state and m grid states produce exactly the state-product upper used in Theorem 6.2. The fixed protocol phase and horizon are supplied by the exogenous public clock; an observation-dependent clock would require additional persistent capacity. A terminal tag challenge emits the stored tag, without consulting a past report. Its $O(\log L)$ output cost is included in the conservative time bound there. Program bits, transient guard bits and the compulsory coarse numerical input are separate resources throughout.

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